



The traditional herbal medicines mixture, Banhasasim-tang, relieves the symptoms of irritable bowel syndrome via modulation of TRPA1, NaV1.5 and NaV1.7 channels

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ABSTRACT

Ethnopharmacological relevance: The cause of irritable bowel syndrome (IBS), a functional gastrointestinal (GI) disorder, remains unclear. Banhasasim-tang (BHSST), a traditional herbal medicines mixture, mainly used to treat GI-related diseases, may have a potential in IBS treatment. IBS is characterized by abdominal pain as the main clinical symptom, which seriously affects the quality of life. *Aim of the study:* We conducted a study to evaluate the effectiveness of BHSST and its mechanisms of action in treating IBS. *Materials and methods:* We evaluated the efficacy of BHSST in a zymosan-induced diarrhea-predominant animal model of IBS. Electrophysiological methods were used to confirm modulation of transient receptor potential (TRP) and voltage-gated Na⁺ (NaV) ion channels, which are associated mechanisms of action. *Results:* Oral administration of BHSST decreased colon length, increased stool scores, and increased colon weight. Weight loss was also minimized without affecting food intake. In mice administered with BHSST, the mucosal thickness was suppressed, making it similar to that of normal mice, and the degree of tumor necrosis factor- α was severely reduced. These effects were similar to those of the anti-inflammatory drug—sulfasalazine—and antidepressant—amitriptyline. Moreover, pain-related behaviors were substantially reduced. Additionally, BHSST inhibited TRPA1, NaV1.5, and NaV1.7 ion channels associated with IBS-mediated visceral hypersensitivity. *Conclusions:* In summary, the findings suggest that BHSST has potential beneficial effects on IBS and diarrhea through the modulation of ion channels.

1. Introduction

A healthy body and mind are crucial for individuals to contribute to the development of society. Health is the most valuable aspect of human life and the foundation of a happy society. It is a great blessing to be able to enjoy an increased lifespan due to the remarkable advancements in medical technology and improved quality of life. However, the problems caused by chronic diseases and aging are on the rise, highlighting the

need to establish a new healthcare paradigm that emphasizes disease prevention and management. In this regard, the benefits of traditional herbal medicines that heal people from a holistic perspective, beyond treating individual diseases, are gaining attention (Huan et al., 2022; Zheng et al., 2022).

Irritable bowel syndrome (IBS) is a functional gastrointestinal (GI) disorder characterized by changes in normal bowel movement. Approximately 11% of the world population suffers from IBS (Canavan et al., 2014). Clinical features of IBS include abdominal pain, abdominal

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A list of abbreviations

IBS	irritable bowel syndrome
BHSST	Banhasasim-tang
GI	gastrointestinal
TRP	transient receptor potential
NaV	voltage-gated Na ⁺
TNF	tumor necrosis factor
I _{TRPV1}	TRPV1 current
I _{TRPV4}	TRPV4 current
I _{TRPA1}	TRPA1 current
I–V	current-voltage
IBS-U	unclassified irritable bowel syndrome
IBS-M	mixed type irritable bowel syndrome
IBS-C	constipation-predominant irritable bowel syndrome
IBS-D	diarrhea-predominant irritable bowel syndrome

distention, and changes in bowel habits, which seriously affect quality of life (Schmulson et al., 2017; Chen et al., 2022). The causes of IBS are multifactorial and may include stress, gut dysbiosis, intestinal dysfunction, visceral hypersensitivity, GI motility disorders, and intestinal infections (Hellstrom, 2019). Although IBS does not cause specific organ changes, it causes discomfort, reduces the quality of life, and increases healthcare burden (Jang et al., 2018; Shih et al., 2002). The zymosan-induced diarrhea-predominant animal model of IBS, made by injecting zymosan into the colon through the anus, is known to exhibit similar symptoms to those of human IBS (Park et al., 2017; Chun et al., 2018). In addition, it is known that clinical symptoms of IBS are controlled by ion channels. Ion channels present throughout the gastrointestinal (GI) tract control the entire digestive process, including secretion, absorption, exercise, and visceral hypersensitivity (Fuentes and Christianson, 2016). Therefore, ion channels have become pharmacological targets in functional GI diseases such as IBS.

Banhasasim-tang (BHSST) is a traditional herbal medicines mixture known as Hange-shashin-to in Japan and anxia-xixin-tang in China. BHSST is mainly used to treat gastrointestinal (GI) diseases such as indigestion, gastric ulcers, laryngopharyngitis, colitis, diarrhea, gastroesophageal reflux disease, and esophageal ulcers due to chronic reflux (Lee et al., 2002; Kawashima et al., 2004; Xu, 2006; F.P. Chen et al., 2009; Shin et al., 2017). Additionally, BHSST is an effective cleaner for treating oral mucosal inflammation and reduces the concentration of PGE2 in oral keratinocytes (Aoyama et al., 2014; Kono et al., 2014). Additionally, BHSST inhibits inflammation-related COX-2 activity (Kase et al., 1998; Kamide et al., 2017), making it an effective anti-inflammatory, antioxidant, and antibacterial agent (Kono et al., 2014; Fukamachi et al., 2015; Hitomi et al., 2016).

According to various clinical and basic studies, traditional medicine, including oriental medicine, has been shown to be effective in the treatment of IBS (Sung et al., 2004; Tang, 2009). The development of safe and efficacious traditional medicines for the treatment of IBS is of great interest to GI physicians (Li et al., 2017). Therefore, the aim of this study was to investigate the efficacy and safety of BHSST as an herbal medicine for IBS treatment and to explore the underlying mechanisms.

2. Materials and methods

2.1. Preparation and qualitative analyses of BHSST

BHSST, a traditional herbal medicines mixture (Pinellia ternata (Thunb.) Makino 0.63 g (tuber), *Scutellaria baicalensis* Georgi 0.85 g (root), *Zingiber officinale* Roscoe (fresh) 0.03 g (rhizome), *Panax ginseng* C.A.Mey. 0.45 g (root), *Glycyrrhiza uralensis* Fisch. ex DC. 0.51 g (rhizome), *Ziziphus jujuba* Mill. 0.34 g (fruit), *Coptis chinensis* Franch.

0.11 g (rhizome) and *Zingiber officinale* Roscoe (dried) 0.28 g (rhizome)) was used what HANKOOKSHINYAK Pharmaceutical Co. LTD. (Powder type, Batch No. 137D213, Nonsan, Republic of Korea) made. The standard compounds used in the BHSST quantitative analyses were experimented by referring to the quantitative indicators prescribed by “The Korean Pharmacopoeia”. *Scutellaria baicalensis* Georgi used baicalin, *Zingiber officinale* Roscoe (dried) used 6-gingerol, *Panax ginseng* C.A. Mey. used ginsenoside Rb1, *Coptis chinensis* Franch. used berberine and *Glycyrrhiza uralensis* Fisch. ex DC. used glycyrrhizic acid, liquiritin, and isoliquiritigenin, as standard compounds. In the case of *Pinellia ternata* (Thunb.) Makino, *Zingiber officinale* Roscoe (fresh), and *Ziziphus jujuba* Mill., the analyses were not conducted because standard compounds for quantification analyses were not prescribed in the “The Korean Pharmacopoeia”. 6-gingerol, glycyrrhizic acid, liquiritin, isoliquiritigenin, berberine, baicalin, and ginsenoside Rb1 were used from Sigma-Aldrich (St Louis, MO, USA). Detection wavelength of each sample for photo diode array (PDA) analysis was as follows: 6-gingerol, 280 nm; glycyrrhizic acid, liquiritin, and isoliquiritigenin, 254 nm; berberine, 345 nm; baicalin, 277 nm; and ginsenoside Rg1, 203 nm. Contents of the 7 marker compounds in BHSST is 6-gingerol 1.06 ± 0.20 ppm, glycyrrhizic acid 10.57 ± 1.30 ppm, liquiritin 3.53 ± 0.15 ppm, isoliquiritigenin 0.73 ± 0.03 ppm, berberine 33.00 ± 0.75 ppm, baicalin 1.11 ± 0.27 ppm and ginsenoside Rb1 1.53 ± 0.13 ppm. Other analyses of BHSST were previously described (Kim et al., 2020). The dosage regimen and experimental design are presented in Fig. 1.

2.2. Animal experiments

Male C57/BL6 mice (Samtako Bio, Osan, Republic of Korea), weighing approximately 21–26 g, were used in this study. The mice were allowed to acclimate for a week before testing. To induce colitis, 0.1 mL zymosan (30 mg/mL; Sigma-Aldrich, St. Louis, MO, USA) was injected into the colon through the anus using a 22-gauge needle. Zymosan or phosphate-buffered saline (PBS) was injected for 3 consecutive days. The mice were divided into six groups: PBS (control), amitriptyline (AMT) (30 mg/kg; Sigma-Aldrich, St. Louis, MO, USA), sulfasalazine (SSZ) (30 mg/kg; Sigma-Aldrich, St. Louis, MO, USA), and BHSST (100, 250, or 500 mg/kg). Naïve mice were administered PBS. The experimental schedule is illustrated in Fig. 1.

2.3. Observation of changes in the colon by zymosan

After removing the stool, colon weight was measured, and colon length was measured as the distance from the end of the cecum to the anus. Stool condition was blindly scored by three researchers using a scale of 0 (normal), 1 (moist), 2 (sticky), and 3 (diarrheal).

2.4. Changes in body weight and food intake

Body weight was measured on days 1, 4, 9, and 12, while food intake was calculated as the sum of the food consumed over the 12-day period.

2.5. Histological examination

For histological examination, tissues were obtained from the large intestine and stained with hematoxylin and eosin (H&E) stain. Cells were identified using a light microscope (Nikon, Tokyo, Japan) at 200 × magnification.

2.6. Enzyme-linked immunosorbent assay (ELISA) measurement

The levels of tumor necrosis factor-alpha (TNF-α) were quantified using an enzyme-linked immunosorbent assay (ELISA) kit from BD Biosciences (San Diego, CA, USA).

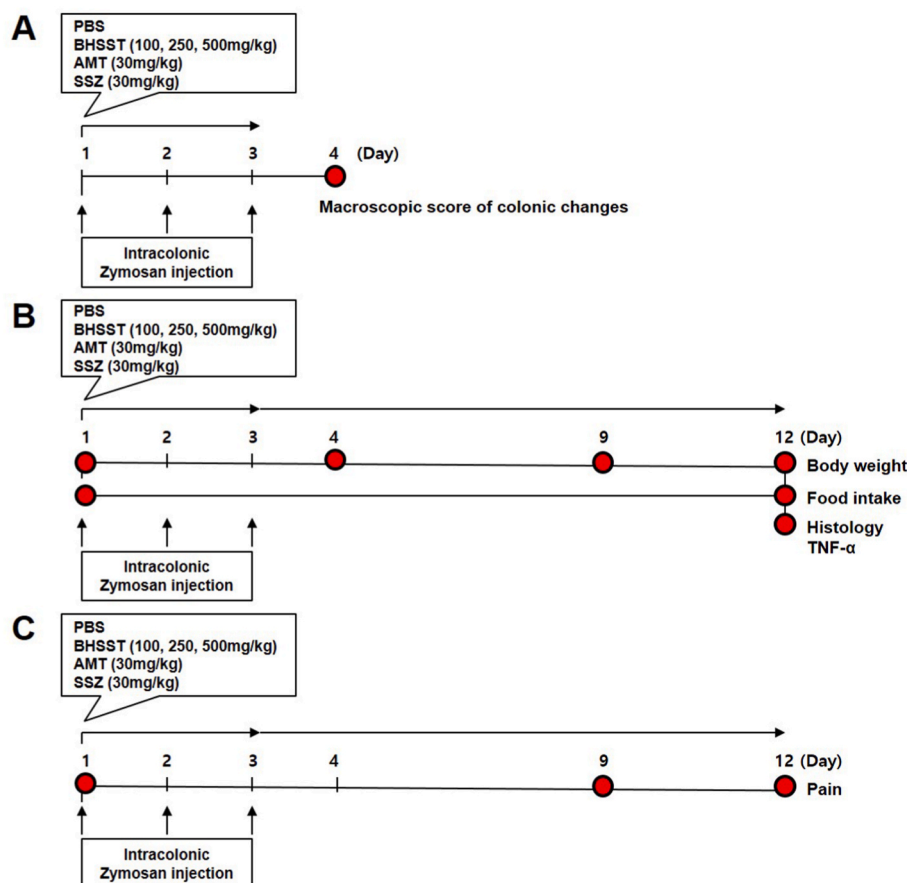


Fig. 1. Experimental design. Mice were administered PBS, BHSST (100, 250, or 500 mg/kg), AMT (30 mg/kg), or SSZ (30 mg/kg) according to the schedule set for the experiment. The experimental design included (A) changes in the colon, (B) body weight, food intake, histological examination, and TNF- α , and (C) pain-related behavior. Naive mice were administered PBS. PBS, phosphate-buffered saline. TNF, tumor necrosis factor.

2.7. Measurement of pain-related behavior

Pain-related behaviors were measured following the methods described in a previous study by Laird et al. (2001). The behaviors to observe were the actions of licking the abdomen, stretching the whole body, putting the abdomen against the floor, or contracting the abdomen in an arched position. Two researchers recorded the number of pain-related behaviors observed within a 10-min timeframe.

2.8. Ethical statement

Animal care and experiments were conducted in accordance with the guidelines of the Institutional Animal Care and Use Committee (IACUC) of Pusan National University (Busan, Republic of Korea; Approval no. PNU-2021-2999) and the National Institutes of Health (NIH) Guide for the Care and Use of Laboratory Animals to ensure ethical treatment of the animals involved.

2.9. Plasmids and transfection

HEK293T cells were used for mammalian cell expression of each ion channel, including transient receptor potential (TRP)V1, TRPV4, TRPA1, and voltage-gated Na⁺ (NaV). Plasmids (0.9 μ g/mL) carrying human TRPV1 (pcDNA5/FRT), TRPV4 (pcDNA3.1), TRPA1 (pcDNA3.1), or NaV1.5/1.7 were transiently transfected into HEK293T cells using the TurboFect transfection reagent (Thermo Fisher Scientific, Waltham, MA, USA) on day 1. To visualize transfected cells, 0.1 μ g/mL of pEGFP-N1 was co-transfected. The cloned hTRPV4 and hTRPA1 genes were provided by Dr. Insuk So (Seoul National University, Seoul,

Republic of Korea). Patch-clamp experiments were conducted 24 h after transfection.

2.10. Electrophysiology

The whole-cell patch clamp technique was used to measure ionic currents in TRP-overexpressing HEK293T cells at room temperature (25 °C). Data were obtained using an Axopatch 200B amplifier (Molecular Devices, San Jose, CA, USA) connected to a Digidata1440A (Molecular Devices, San Jose, CA, USA) and organized using pCLAMP 10.7 (Molecular Devices, San Jose, CA, USA). The pipette resistance was 2–3 M Ω when filled with the internal solution. The transfected cells were transferred to a bath chamber mounted on an inverted microscope (Ti-U, Nikon, Japan). Bath solutions were applied using a gravity-driven local perfusion system at a perfusion speed of approximately 10 mL/min. All voltage and current trace data were analyzed using Clampfit 10.7 (Molecular Devices, San Jose, CA, USA). For the NaV channels, the amount of current generated by changing the membrane voltage from –120 mV to 0 mV was measured after fixing the cell membrane voltage at –120 mV. The compositions of the external and internal solutions are listed in Table 1.

2.11. Statistical analysis

The results were presented as means $\pm$ SEMs, and Tukey's multiple comparison test was used for the analysis of variance (ANOVA) using GraphPad Prism 6. Statistical significance was set at p values < 0.05.

Table 1

The composition of external and internal solutions.

TRPV1			pH
Internal Solution	140 CsCl, 10 NaCl, 5 EGTA, 10 HEPES, 3 Mg-ATP		7.2 (adjusted with CsOH)
External Solution	140 NaCl, 4 KCl, 10 HEPES, 1 EGTA, 1 MgCl ₂ , 5 glucose		7.4 (adjusted with NaOH)
TRPV4			pH
Internal Solution	120 CsOH, 120 L-glutamic acid, 20 CsCl, 5 HEPES, 1 MgCl ₂ , 10 EGTA, 3 Mg-ATP		7.2 (adjusted with CsOH)
External Solution	145 NaCl, 3.6 KCl, 10 HEPES, 5 glucose, 1 MgCl ₂ , 1.3 CaCl ₂		7.4 (adjusted with NaOH)
TRPA1			pH
Internal Solution	150 CsCl, 10 HEPES, 1 MgCl ₂ , 20 BAPTA		7.2 (adjusted with CsOH)
External Solution	150 NaCl, 10 HEPES, 2 CaCl ₂ , 1 MgCl ₂		7.4 (adjusted with NaOH)
NaV			pH
Internal Solution	100 CsF, 45 CsCl, 5 EGTA, 10 HEPES, 5 NaCl		7.2 (adjusted with CsOH)
External Solution	138 NaCl, 4 KCl, 1 MgCl ₂ , 1.8 CaCl ₂ , 10 HEPES, 5.6 D-Glucose		7.45 (adjusted with NaOH)

3. Results

3.1. Effects of BHSST on the zymosan-induced colon changes

We evaluated colon length, colon weight, and stool condition in zymosan-injected colons after BHSST administration. The antidepressant drug AMT and anti-inflammatory drug SSZ, clinically used to treat patients with IBS, were used as positive controls (Morgan et al., 2005; Pruzanski et al., 1997). Zymosan administration reduced colon length compared to naive mice, indicating colitis induction. However, BHSST administration restored the colon length to normal [7.3 ± 0.6 cm at naive, 5.7 ± 0.1 cm ($p < 0.01$) at control, 7.1 ± 0.6 cm ($p < 0.01$) at BHSST 100 mg/kg, 7.0 ± 0.6 cm ($p < 0.01$) at BHSST 250 mg/kg, 7.2 ± 0.5 cm ($p < 0.01$) at BHSST 500 mg/kg, 7.4 ± 0.5 cm ($p < 0.01$) at SSZ, and 7.2 ± 0.6 cm ($p < 0.01$) at AMT; Fig. 2A]. Furthermore, colon

weight significantly increased compared to that in the naive group, indicating the occurrence of colitis. However, BHSST administration restored colon weight to normal [0.25 ± 0.03 g at naive, 0.35 ± 0.01 g ($p < 0.01$) at control, 0.26 ± 0.01 g ($p < 0.01$) at BHSST 100 mg/kg, 0.24 ± 0.02 g ($p < 0.01$) at BHSST 250 mg/kg, 0.25 ± 0.01 g ($p < 0.01$) at BHSST 500 mg/kg, 0.26 ± 0.04 g ($p < 0.05$) at SSZ, and 0.28 ± 0.01 g ($p < 0.01$) at AMT; Fig. 2B]. Feces in naive mice were dark brown color and hard mass, whereas zymosan-injected mice showed watery and sticky diarrhea with bright brown or yellow color. As a result, stool score was higher in zymosan-treated mice than in naive mice. However, BHSST administration significantly reduced the stool score compared to that in control mice [0.08 ± 0.02 at naive, 2.50 ± 0.16 ($p < 0.01$) at control, 1.42 ± 0.09 ($p < 0.01$) at BHSST 100 mg/kg, 1.22 ± 0.09 ($p < 0.01$) at BHSST 250 mg/kg, 0.51 ± 0.09 ($p < 0.01$) at BHSST 500 mg/kg, 0.09 ± 0.02 ($p < 0.01$) at SSZ, and 0.08 ± 0.02 g ($p < 0.01$) at AMT; Fig. 2C]. These results suggest that BHSST administration inhibits zymosan-induced colitis and diarrhea.

3.2. Effects of BHSST on body weight and food intake

Following zymosan administration, control mice experienced weight loss on days 4, 9, and 12 compared to naive mice. However, oral administration of BHSST inhibited weight loss on days 4 [25.1 ± 0.5 g at naive, 23.5 ± 0.8 g ($p < 0.01$) at control, 24.2 ± 1.0 g ($p < 0.05$) at BHSST 100 mg/kg, 23.9 ± 0.9 g at BHSST 250 mg/kg, 24.1 ± 1.6 g at BHSST 500 mg/kg; Fig. 2A], 9 [26.2 ± 1.1 g at naive, 24.1 ± 0.3 g ($p < 0.01$) at control, 25.1 ± 0.9 g at BHSST 100 mg/kg, 25.5 ± 0.5 g ($p < 0.01$) at BHSST 250 mg/kg, 25.5 ± 1.8 g at BHSST 500 mg/kg; Fig. 3A], and 12 [27.1 ± 1.4 g at naive, 24.7 ± 0.5 g ($p < 0.01$) at control, 25.7 ± 0.9 g at BHSST 100 mg/kg, 25.7 ± 0.3 g ($p < 0.01$) at BHSST 250 mg/kg, 25.7 ± 1.7 g at BHSST 500 mg/kg; Fig. 3A]. Administration of AMT and SSZ significantly increased body weight on days 9 [25.3 ± 0.4 g ($p < 0.01$) at SSZ, and 25.5 ± 0.7 g ($p < 0.01$) at AMT] and 12 [25.9 ± 0.4 g ($p < 0.01$) at SSZ, and 26.2 ± 0.7 g ($p < 0.01$) at AMT] (Fig. 3A). There was no difference in food intake between groups (Fig. 3B). These results suggest that BHSST does not affect food intake or prevent weight loss in

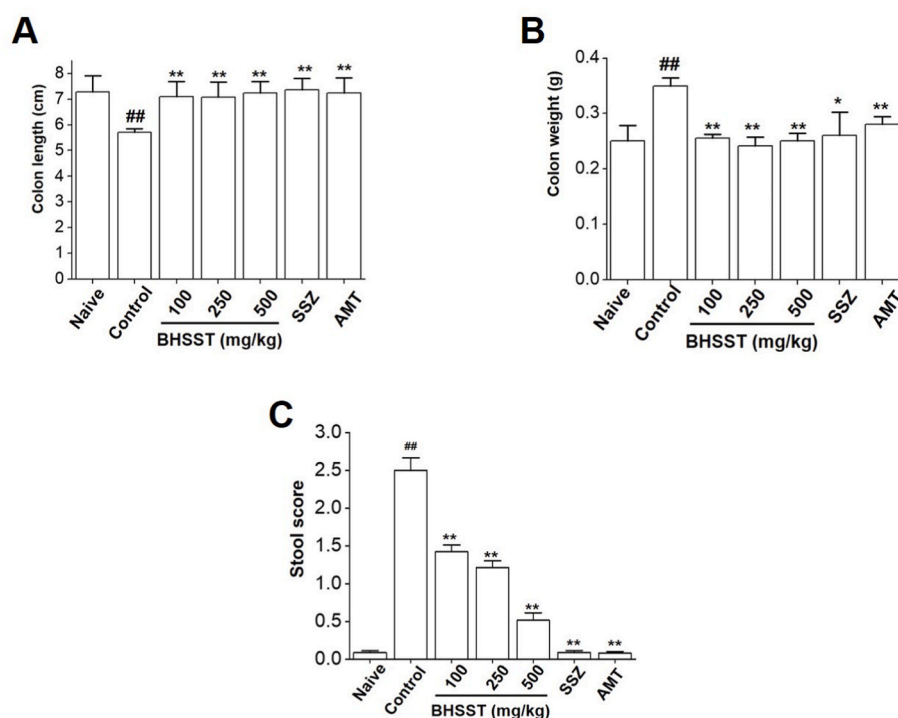


Fig. 2. Effects of BHSST on colon changes induced by zymosan injection. (A) Colon length. (B) Colon weight. (C) Stool score. Bars represent mean ± SE. ## $p < 0.01$ vs. naive. * $p < 0.05$, ** $p < 0.01$ vs. control.

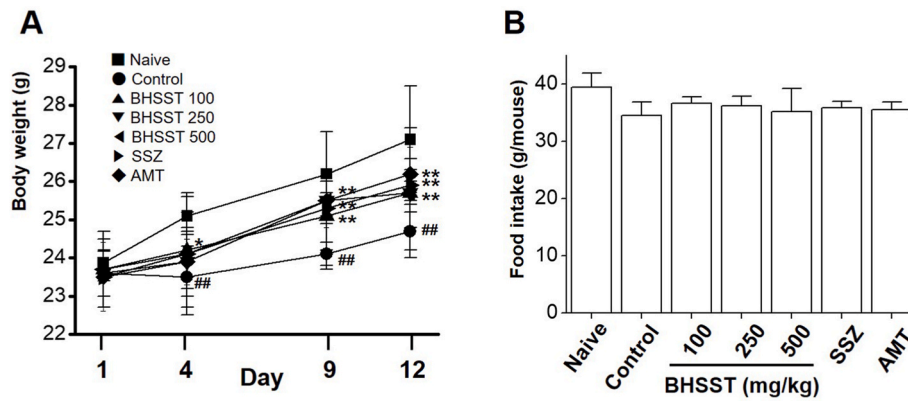


Fig. 3. Effects of BHSST on body weight and food intake in IBS induced by zymosan injection. (A) Body weight. (B) Food intake. Bars represent mean $\pm$ SE. $^{##}p < 0.01$ vs. naive. $^{*}p < 0.05$, $^{**}p < 0.01$ vs. control.

zymosan-induced IBS in mice.

3.3. Effects of BHSST on zymosan-induced colitis

Histological changes in the colon were confirmed by hematoxylin and eosin (H&E) staining. As previously reported, inflammation occurred in the colons of zymosan-injected mice compared to that in naive mice (Coutinho et al., 1996; Park et al., 2017). However, H&E staining showed a substantially reduced colon wall thickness in BHSST-treated mice (Fig. 4A and B). In addition, the level of TNF- α was remarkably increased in zymosan-treated mice, but BHSST inhibited this increase and almost returned it to normal levels (Fig. 4C). These findings suggest that BHSST may have an inhibitory effect on gastrointestinal inflammation in zymosan-induced IBS.

3.4. Effects of BHSST on pain-related behaviors

On day 1, no pain-related behaviors were observed in zymosan-

treated mice compared with naive mice (Fig. 5A). However, on day 6, zymosan-treated mice showed pain-related behaviors, such as licking the abdomen, attaching the abdomen to the floor, stretching the entire body, and grabbing the abdomen. BHSST, SSZ, or AMT reduced pain-related behaviors on day 7 [28.2 ± 2.2 at naive, 38.8 ± 2.9 ($p < 0.01$) at control, 35.6 ± 3.0 at BHSST 100 mg/kg, 30.0 ± 3.2 ($p < 0.05$) at BHSST 250 mg/kg, 28.8 ± 3.7 ($p < 0.05$) at BHSST 500 mg/kg, 30.4 ± 3.6 cm ($p < 0.05$) at SSZ, and 29.8 ± 3.8 cm ($p < 0.05$) at AMT; Fig. 5B]. These pain-related actions did not persist until day 12 (Fig. 5C). These results suggest that BHSST suppresses pain in zymosan-induced IBS mice.

3.5. Effects of BHSST on TRP channel currents

To determine whether BHSST regulates hTRPA1 channels, we performed whole-cell electrophysiological recordings on mock-transfected cells and TRPA1-overexpressed HEK293T cells. We obtained the current-voltage (I-V) curve at each stable time-point using a ramp pulse

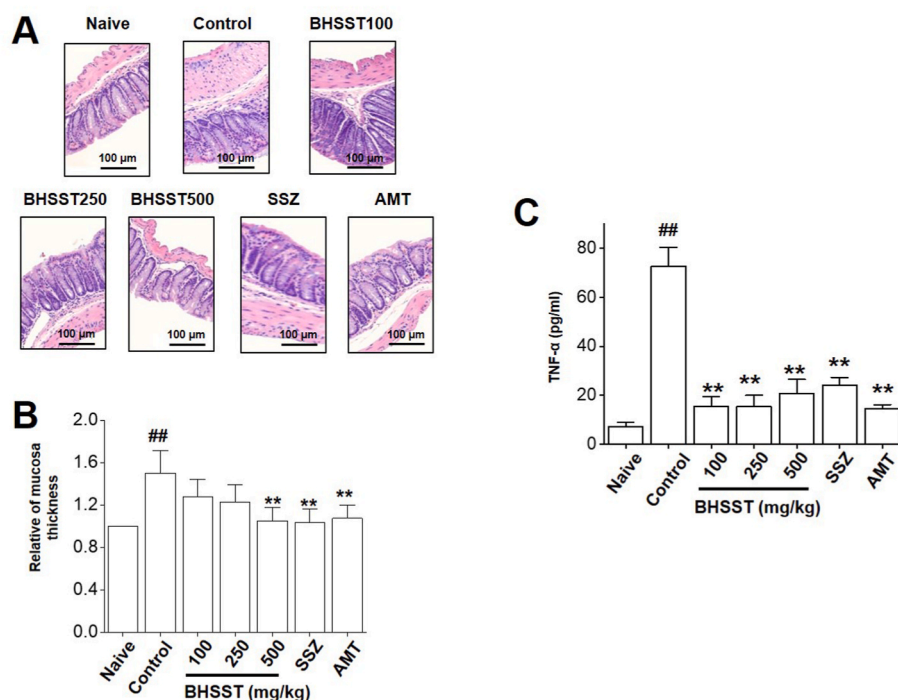


Fig. 4. Effects of BHSST on mucous thickness and TNF- α levels in zymosan-induced IBS. (A) Hematoxylin and eosin staining. (B) Relative mucous thickness. (C) TNF- α levels. Bars represent mean $\pm$ SE. $^{##}p < 0.01$ vs. naive. $^{**}p < 0.01$ vs. control.

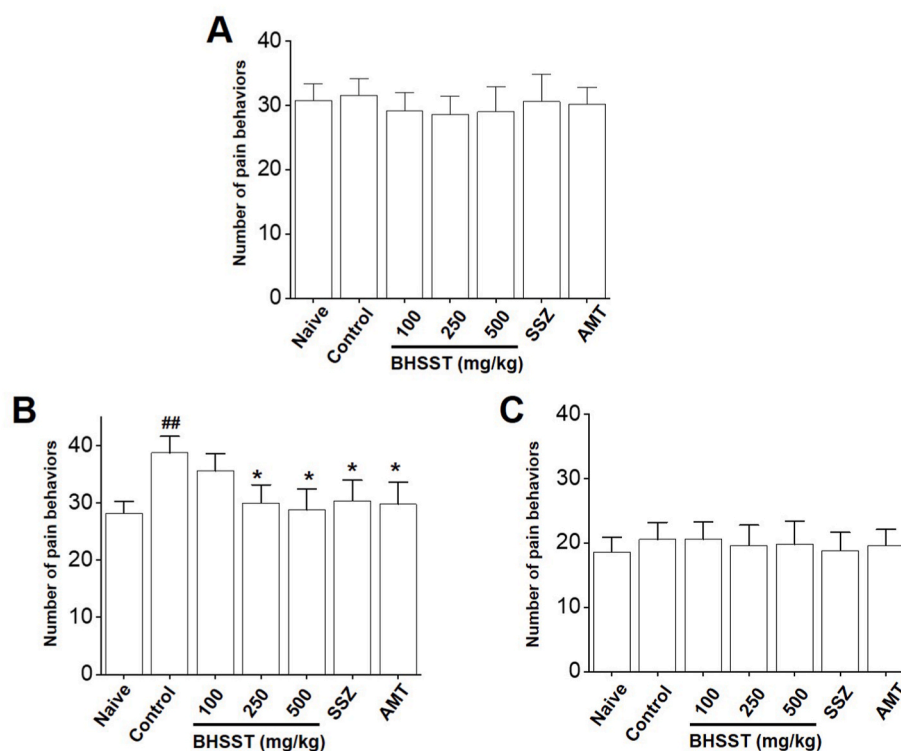


Fig. 5. Effects of BHSST on pain-related behaviors in zymosan-induced IBS. The number of pain-related behaviors was recorded on days (A) 1, (B) 9, and (C) 12. Bars represent mean $\pm$ SE. ^{##} $p < 0.01$ vs. naive. * $p < 0.05$ vs. control.

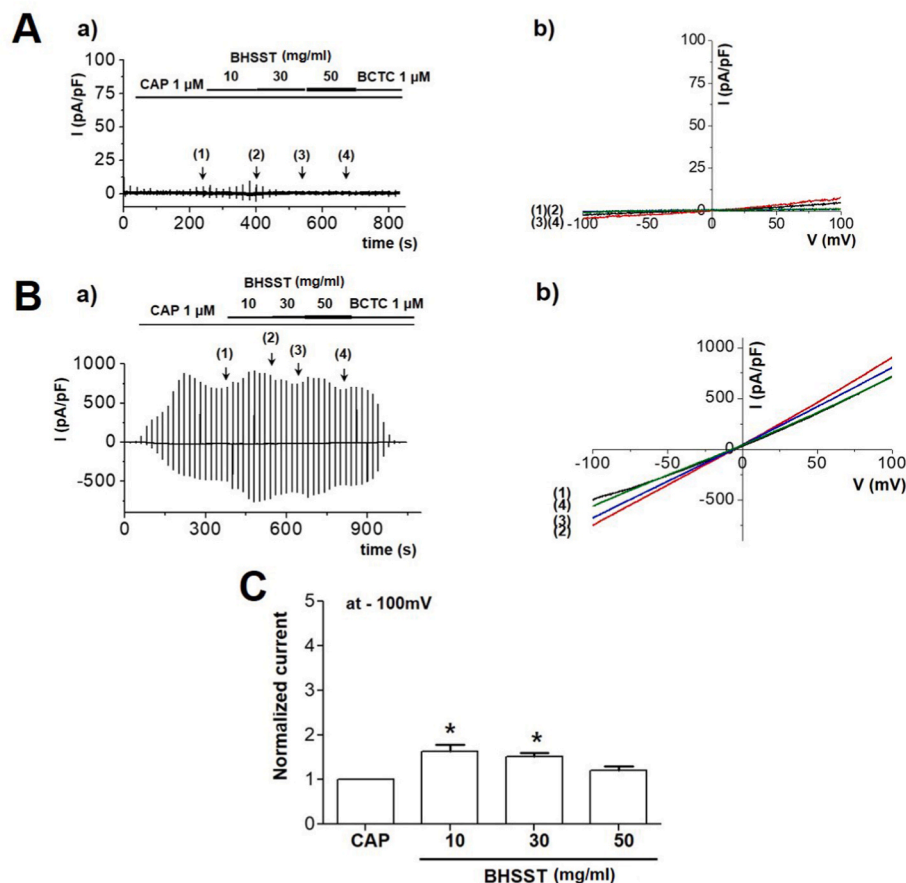


Fig. 6. Pharmacological effects of BHSST on IBS-related TRPV1 currents in HEK293T cells with TRPV1 overexpression. (A) (a) Representative trace showing the effects of BHSST (10, 30, and 50 mg/mL) on mock-overexpressed HEK293T cells. (b) Representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on mock-overexpressed HEK293T cells. (B) (a) Representative trace showing the effects of BHSST (10, 30, and 50 mg/mL) on TRPV1-overexpressed HEK293T cells. (b) Representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on TRPV1-overexpressed HEK293T cells. (C) Statistical analysis of normalized fold changes of I_{TRPV1} by BHSST (10, 30, and 50 mg/mL) at -100 mV, where the inward I_{TRPV1} using capsaicin was set to 1. * $p < 0.05$ vs. control.

from -100 mV to 100 mV under control (1), BHSST 10 mg/mL (2), BHSST 30 mg/mL (3), and BHSST 50 mg/mL (4) conditions, and following treatment with the selective TRPV1 inhibitor, BCTC (Heber et al., 2020). The TRPV1 current (I_{TRPV1}) is activated by capsaicin (CAP) (C. Zhang et al., 2020). I_{TRPV1} was not activated, and perfusing CAP into the bath had no effect on mock-transfected cells (Fig. 6A). However, after bath application of 1 μ M CAP, I_{TRPV1} was evoked, and application of 10 mg/mL ($p < 0.05$) or 30 mg/mL ($p < 0.05$) of BHSST produced I_{TRPV1} increases at all potentials (Fig. 6B). However, application of 50 mg/mL BHSST had no effect on I_{TRPV1} expression (Fig. 6B). Fig. 6C shows the average current amplitudes at each concentration at -100 mV. To determine whether BHSST regulates hTRPV4 channels, we performed whole-cell electrophysiological analyses on mock-transfected cells and TRPV4-overexpressed HEK293T cells. We obtained the current-voltage (I-V) curve at each stable time-point using a ramp pulse from -100 mV to 100 mV under control (1), BHSST 10 mg/mL (2), BHSST 30 mg/mL (3), and BHSST 50 mg/mL (4) conditions, and following treatment with the selective TRPV4 inhibitor, ruthenium red (RR) (Ueda et al., 2011). TRPV4 current (I_{TRPV4}) is activated by GSK101A (Jin et al., 2011). I_{TRPV4} was not activated, and perfusing GSK101A into the bath had no effect on mock-transfected cells (Fig. 7A). However, after bath application of 300 nM GSK101A, I_{TRPV4} was evoked, and application of 10 , 30 ($p < 0.05$), or 50 mg/mL BHSST produced I_{TRPV4} increases at both positive and negative potentials (Fig. 7B). Fig. 7C shows the average current amplitudes at each concentration at -100 mV. To determine whether BHSST regulates hTRPA1 channels, we used a whole-cell electrophysiological method on mock-transfected cells and TRPA1 overexpressed HEK293T cells. We obtained the current-voltage (I-V) curve at each stable time-point using a ramp pulse

from -100 mV to 100 mV at control (1), BHSST 10 mg/mL (2), BHSST 30 mg/mL (3), and BHSST 50 mg/mL (4), and treatment with the selective TRPA1 inhibitor, A967079 (Gyamfi et al., 2020). It is known that I_{TRPA1} is activated by AITC (Sandor et al., 2016). I_{TRPA1} was not activated, and perfusing AITC into the bath had no effect on the mock-transfected cells (Fig. 8A). However, after bath application of 100 μ M AITC, I_{TRPA1} was evoked, and application of 10 ($p < 0.01$), 30 ($p < 0.01$), or 50 mg/mL ($p < 0.01$) of BHSST inhibited I_{TRPV4} (Fig. 8B). Fig. 8C shows the average current amplitudes at each concentration at -100 mV. These results suggest that TRPA1 may be the main factor regulating the inhibition of IBS-induced visceral pain.

3.6. Effects of BHSST on NaV1.5 or 1.7 channel currents

The effects of BHSST on NaV1.5 or 1.7 channel currents expressed in HEK293T cells were examined by recording whole-cell patch-clamp recordings. The currents evoked rapidly decaying transient inward NaV1.5 or 1.7 channel currents. As depicted in Fig. 9, the addition of BHSST resulted in a significant reduction in the peak inward current. BHSST inhibited the peak currents by $73.7 \pm 6.6\%$ at 10 mg/mL ($p < 0.01$), $41.1 \pm 8.6\%$ ($p < 0.01$) at 30 mg/mL, and $19.9 \pm 1.7\%$ ($p < 0.01$) at 50 mg/mL, with an IC_{50} of 21.4 mg/mL in NaV1.5 channels. Furthermore, as demonstrated in Fig. 10, the addition of BHSST significantly reduced the peak inward current. BHSST inhibited the peak currents by $79.1 \pm 5.0\%$ at 10 mg/mL, $42.2 \pm 8.6\%$ ($p < 0.01$) at 30 mg/mL, and $15.8 \pm 4.6\%$ ($p < 0.01$) at 50 mg/mL, with an IC_{50} of 22.7 mg/mL in NaV1.7 channels. These findings suggest that NaV1.5 or 1.7 current may be factors regulating the inhibition of IBS-induced visceral pain.

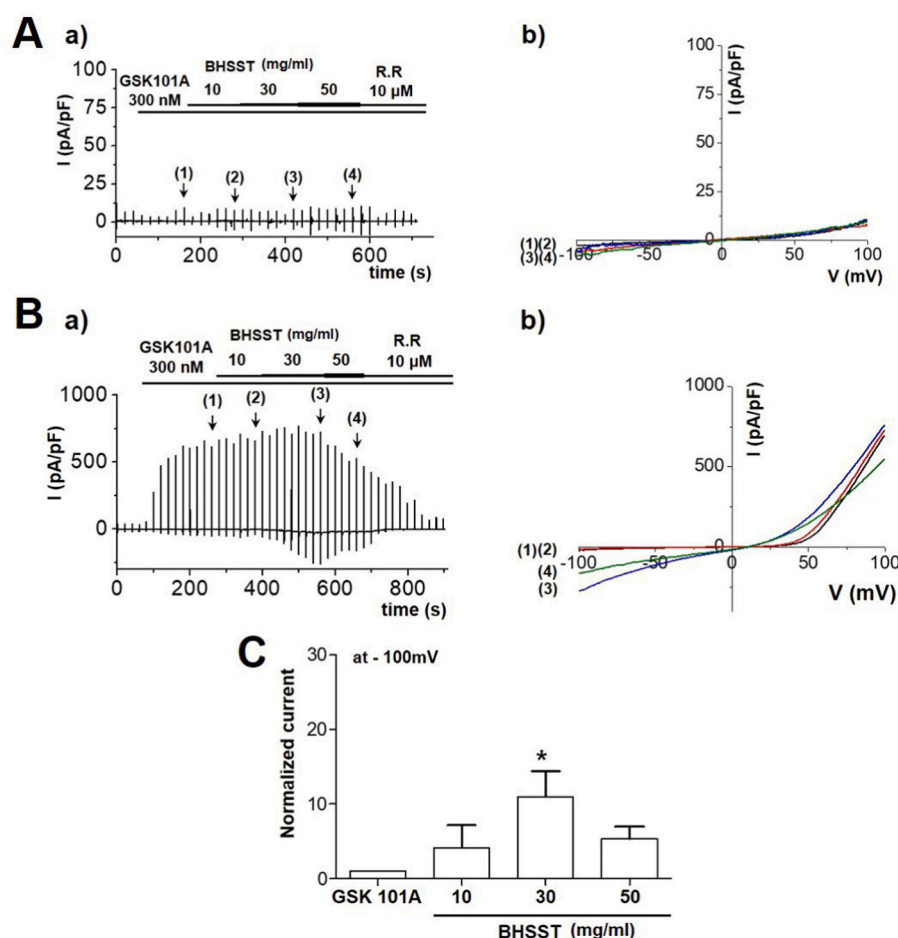


Fig. 7. Pharmacological effects of BHSST on IBS-related TRPV4 currents in HEK293T cells with TRPV4 overexpression. (A) (a) Representative plot and (b) representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on mock-overexpressed HEK293T cells. (B) (a) Representative plot and (b) representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on HEK293T cells with TRPV4 overexpression. (C) Summary of fold changes in I_{TRPV4} following treatment with BHSST (10, 30, and 50 mg/mL) at -100 mV, where the inward I_{TRPV4} evoked by GSK101A was set to 1. * $p < 0.05$ vs. control.

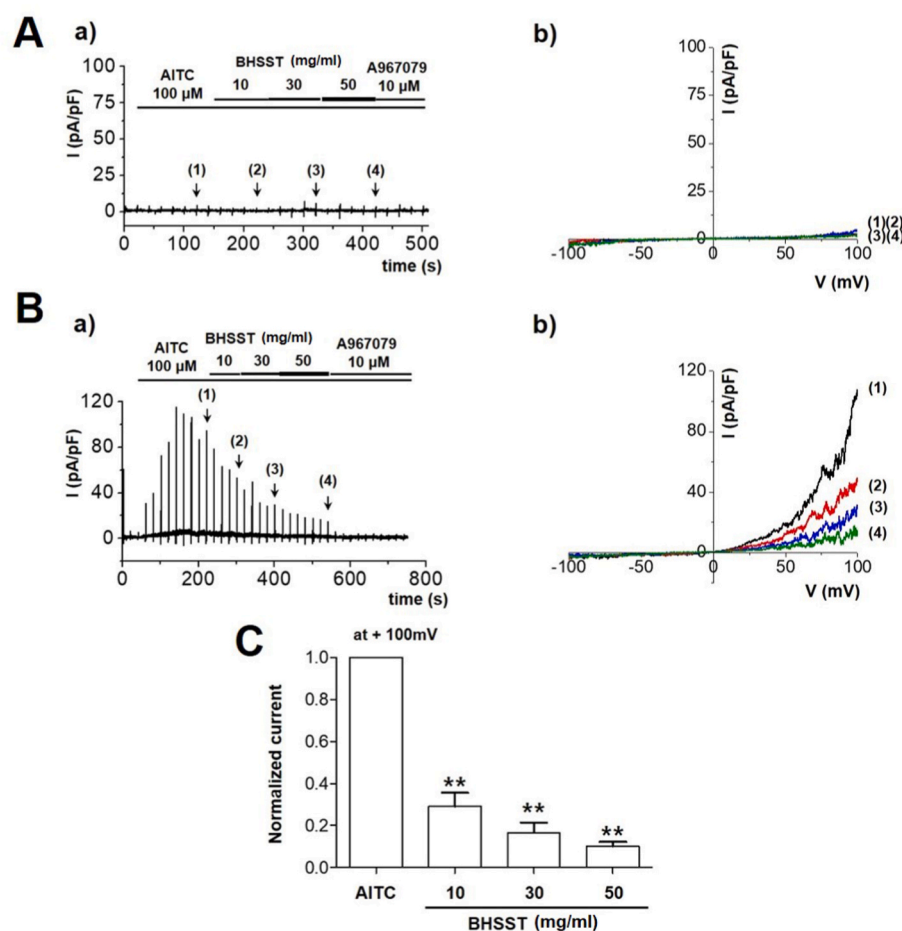


Fig. 8. Pharmacological effects of BHSST on IBS-related TRPA1 currents in HEK293T cells with TRPA1 overexpression. (A) (a) Representative plot and (b) representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on mock-overexpressed HEK293T cells. (B) (a) Representative plot and (b) representative I-V curve showing the effects of BHSST (10, 30, and 50 mg/mL) on HEK293T cells with TRPA1 overexpression. (C) Statistical analysis of normalized fold changes of I_{TRPA1} by BHSST (10, 30, and 50 mg/mL) at +100 mV, where the inward I_{TRPA1} using AITC was set to 1. ** $p < 0.01$ vs. control.

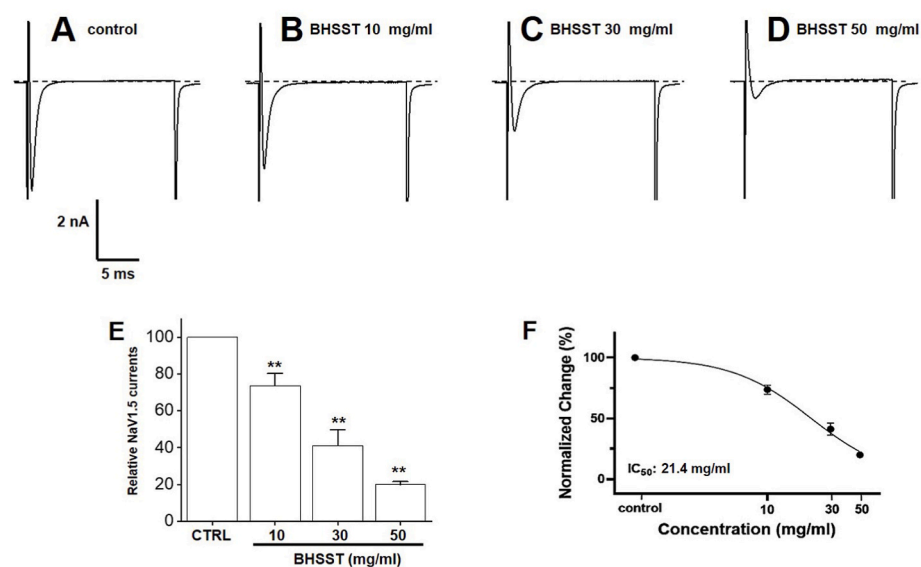


Fig. 9. Pharmacological effects of BHSST on NaV1.5 currents in HEK293T cells overexpressing NaV1.5, an ion channel associated with IBS. Representative plot showing the effects of BHSST at different concentrations (A) control, (B) 10, (C) 30, and (D) 50 mg/mL on NaV1.5 currents in HEK293T cells. (E) Statistical analysis of normalized fold changes of NaV1.5 currents by BHSST (10, 30, and 50 mg/mL). (F) Concentration-dependent inhibition of NaV1.5 by BHSST. The estimated IC_{50} was 21.4 mg/mL. The inward NaV1.5 currents in control were set to 100%. ** $p < 0.01$ vs. control.

4. Discussion

IBS is a chronic functional GI disorder that causes repeated abdominal pain and changes in bowel habits for at least six months (Canavan et al., 2014). Although IBS is not life-threatening, it causes pain, discomfort, and loss of motivation in daily life for patients (Schmulson

et al., 2017). Frequent bowel movements and abdominal pain can also interfere with social activities (Chen et al., 2022). The pathological mechanism of IBS is not clearly understood, but it is known to involve psychology, the central nervous system, neuroendocrine responses, GI movement, and abdominal hypersensitivity (Chaudhury et al., 2015). Depending on clinical features, IBS can be divided into unclassified

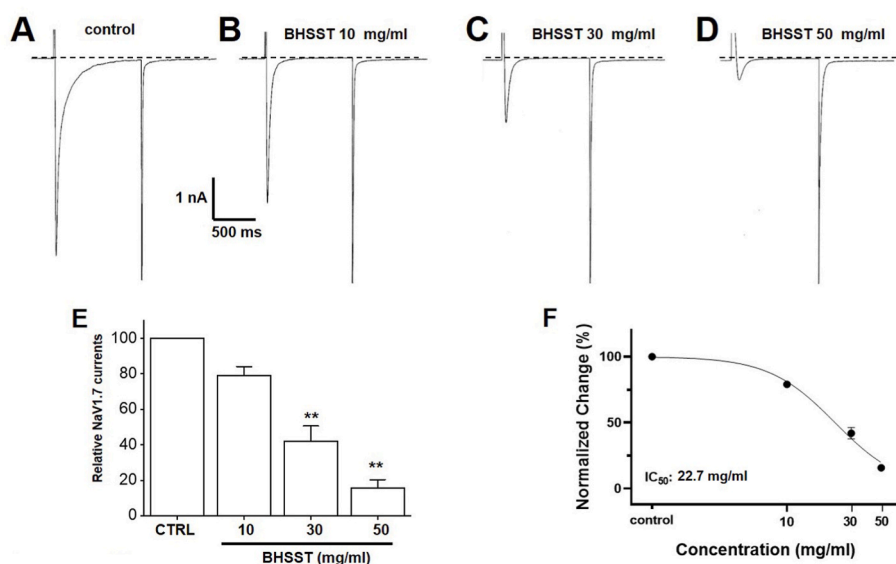


Fig. 10. Pharmacological effects of BHSST on NaV1.7 currents in HEK293T cells overexpressing NaV1.7, another ion channel associated with IBS. Representative plot showing the effects of BHSST at different concentrations (A) control, (B) 10, (C) 30, and (D) 50 mg/mL on NaV1.7 currents in HEK293T cells. (E) Statistical analysis of normalized fold changes of NaV1.7 currents by BHSST (10, 30, and 50 mg/mL). (F) Concentration-dependent inhibition of NaV1.7 by BHSST. The estimated IC₅₀ was 22.7 mg/mL. The inward NaV1.7 currents in control were set to 100%. **p < 0.01 vs. control.

(IBS-U), mixed type (IBS-M), constipation-predominant (IBS-C), and diarrhea-predominant (IBS-D) (Drossman and Hasler, 2016; Lee and Lee, 2014). Among these, IBS-D is the most common. The type of IBS may also be determined based on intestinal microorganisms, with IBS-C being the most relevant type (Mohammadzadeh et al., 2019). The human intestine is a fairly complex structure, and various microorganisms such as bacteria, viruses, and eukaryotes etc. (Marchesi, 2010; Flayyaz et al., 2021). Gut microbiota play many key roles in maintaining health. Mainly, they are involved in metabolism such as amino acid and vitamin production, prevent the growth of germs, and maintain intestinal structures. (Akash et al., 2019). The most common bacterial phyla are *Firmicutes* and *Bacteroidetes* (Hugon et al., 2015; Akash et al., 2019). Recent studies have shown that the human gut is likely to be affected by environmental factors including stress, eating habits, and lifestyle throughout an individual's life span. Furthermore, recent studies suggest that gut microbiome is an important key factor in IBS development (Rothschild et al., 2018). Short-chain fatty acids (SCFA), such as acetic acid and butyrate, are also produced by microorganisms in the large intestine, which can affect intestinal movement as well as intestinal permeability. According to a study by Pozuelo et al. (2015), IBS-D and IBS-M have a low content of bacteria that produce butyrate. Bacteria that produce butyrate are known to improve intestinal barrier function, so in patients with IBS-D and IBS-M, it is known that reducing the amount of these bacteria decreases intestinal permeability. In addition, IBS patients who received no treatment had little metanobacteria (Pozuelo et al., 2015), and methane production was low in IBS-D and high in IBS-C (Jahng et al., 2012; Tap et al., 2017). Therefore, IBS can be considered a multifactorial disease caused by a combination of various factors in the central and peripheral regions.

BHSST is a well-known traditional medicinal treatment that is widely used to treat digestive diseases such as vomiting and swelling of the upper abdomen (Kawashima et al., 2006). BHSST controls the stomach and removes reflux of "qi," and is widely used to ameliorate congestion and fullness in the GI tract (Kawashima et al., 2006). BHSST is also used to treat acute and chronic diarrhea and GI inflammation and shows more effective treatment efficacy when the function of the GI tract is reduced (Kase et al., 1996; Takasuna et al., 1995). Additionally, BHSST is used to treat diarrhea, side effects of anticancer drugs, prevent alcohol-induced gastric ulcers, and heal damaged gastric mucosal tissue (Shin et al., 2017). In this study, BHSST decreased colon length and stool score and increased colon weight in a zymosan-induced diarrhea-type IBS model (Fig. 2). The amount of food intake remained unchanged, and the body weight of the mice increased (Fig. 3). The thickness of the intestinal

mucosa was reduced, and the level of the inflammatory marker TNF- α was reduced (Fig. 4). Moreover, BHSST reduced pain-related behaviors (Fig. 5). BHSST upregulated TRPV1 (Fig. 6) and TRPV4 (Fig. 7), ion channels associated with visceral hypersensitivity, and decreased TRPA1 (Fig. 8). Additionally, BHSST downregulated NaV1.5 (Fig. 9) and NaV1.7 channels (Fig. 10). Therefore, this study confirmed that BHSST is effective in the treatment of IBS via the modulation of TRPA1, NaV1.5, and NaV1.7 channels.

Among the ion channels, transient receptor potential (TRP) channels are widely distributed in mammalian tissues (Gunthorpe et al., 2002; Vriens et al., 2004). TRP channels recognize temperature, harmful substances, and pain (Zielińska et al., 2015) and play a role in the immunity and pathogenesis of IBS (Zielińska et al., 2015). TRPV1, TRPV4, and TRPA1 are involved in GI hypersensitivity (Beckers et al., 2017). TRPV1 plays an important role in mechanosensation and inflammatory distress in the colorectum and, therefore, is a pharmacological target for IBS (Stucky et al., 2009). TRPV4 is involved in the visceral hypersensitivity of the large intestine (Cenac et al., 2015). TRPA1, which is associated with PAR2 activation in the colon tissues, is an effector protease associated with visceral hypersensitivity (Yu et al., 2016). NaV channels are emerging drug targets for pain suppression (Emery et al., 2016). Among the NaV channels, NaV1.5, the predominant GI NaV channel (Strege et al., 2003), and NaV1.7 have received great attention in the development of new pain inhibitors (Emery et al., 2016). Inhibition of NaV1.5, associated with diarrhea and pain, is associated with the recovery of normal GI motility in IBS (Nash and Nash, 2008; Wang et al., 2018), and NaV1.7 is becoming a treatment option for patients with IBS-D (Verne et al., 2005). In this study, BHSST increased the expression of TRPV1 (Fig. 6) and TRPV4 (Fig. 7) and decreased the expression of TRPA1 (Fig. 8). Additionally, BHSST reduced the response of NaV1.5 (Fig. 9) and NaV1.7 (Fig. 10). Therefore, the reduction in pain associated with IBS inhibition by BHSST can be attributed to TRPA1, NaV1.5, and NaV1.7. Like TRP and NaV channels, ion channels often change their expression level and function by IBS, and thus become pharmacological targets of IBS. Among them, it was found that the expression level of T-type voltage-gated calcium channel (Cav) 3.2 increased in human IBS patients (Fuentes and Christianson, 2016), so it is necessary to check the expression level and the possibility of BHSST regulation of Cav3.2 in mice. In addition, future studies should investigate the efficacy and potential roles of other ion channels related to visceral hypersensitivity in the GI tract. In addition, this study is a study with mice, and we plan to confirm the possibility of treating IBS in humans in the future. So far, many papers have published that BHSST is effective in IBS (Park et al.,

2010, 2013; Jeon et al., 2019; Kim et al., 2021). We will continue to check the clinical efficacy by referring to these papers.

For BHSST quantitative analyses, 6-gingerol, glycyrrhizinic acid, liquiritin, isoliquiritigenin, berberine, baicalin, and ginsenoside Rb1 were used as standard compounds. 6-gingerol is the most abundant component of *Zingiber officinale* Roscoe (dried) and is absorbed well in the GI tract (Durojaye et al., 2019). It is known to have anti-cancer, anti-inflammatory, anti-oxidant, and anti-nausea, which are important for maintaining health (Promdam and Panichayupakaranant, 2022). It is not well known how it acts on IBS, but it is known to have a function of suppressing ulcerative colitis in the intestine (Sheng et al., 2020). Also, it is said to function as an agonist for TRPV1 and antagonist for TRPA1 and inhibit NaV1.8 (Hitomi et al., 2017; Morera et al., 2012). Glycyrrhizinic acid is one of the main components of *Glycyrrhiza uralensis* Fisch. ex DC., and its anti-cancer, anti-inflammatory, antiviral, antibacterial, and anti-allergic effects have been known (Chen et al., 2020). Because of these pharmacological characteristics, it is effective in treating various intestinal inflammatory diseases (Chen et al., 2019). However, the effect in IBS has not well known yet. The efficacy of ion channels is known to inhibit delayed rectifier K⁺ and human ether-a-go-go related gene (HERG) K⁺ channels, but the efficacy on TRP channels or NaV 1.5 and 1.7 has not been known yet (Wu et al., 2013). Liquiritin is a flavonoid derived from *Glycyrrhiza uralensis* Fisch. ex DC. and is widely used as a traditional medicine to boost energy, remove heat, and eliminate toxins in the body (Qin et al., 2022). It has the characteristics of passing through the blood-brain-barrier and has anti-Alzheimer, anti-depression, anti-cancer, anti-inflammatory, heart, liver and skin protection effects (Qin et al., 2022). The efficacy of IBS has been unknown, but it has been known to be involved in the control of bacteria in the GI tract (Zhang et al., 2014). Looking at the efficacy of ion channels, TRPA1 and TRPV1 channels are inhibited (Liu et al., 2020). Isoliquiritigenin is one of the important bioactive ingredients of *Glycyrrhiza uralensis* Fisch. ex DC. and commonly used in food and traditional medicine as a food additive and cancer or antibiotic treatment (Peng et al., 2015). It also has anti-inflammatory, anti-microbial, antioxidant, immune control functions, liver and heart protection effects, so it has been used to treat and prevent various diseases (Peng et al., 2015). It has a function of controlling GI motility by controlling muscarinic receptors and calcium channels in the GI tract, and also has a function of suppressing MAPK mechanisms to relieve colitis caused by dextran sodium sulfate (DSS) (G. Chen et al., 2009; Choi et al., 2016). In addition, it is known that this has no effect on the TRPV1 channels, but inhibits the TRPA1 and various NaV channels to reduce pain (Miyamura et al., 2021; Ramalingam et al., 2018). Berberine is a widely used traditional medicine characterized by antibacterial and anti-inflammatory as a component of *Coptis chinensis* Franch. (Ai et al., 2021). It has many pharmacological characteristics and therefore, it can be used to treat various diseases, especially for vascular diseases (Ai et al., 2021). It has the effect of reducing inflammation of the intestinal mucosa, restoring intestinal motility, and suppressing abdominal pain in IBS (Yu et al., 2019). It also inhibits TRPV1, TRPV4 and TRPA1 channels (Meng et al., 2022; Wang et al., 2015; Zou et al., 2022). Baicalin is one of the important bioactive flavonoids of *Scutellaria baicalensis* Georgi and has antiviral, antibacterial, and anti-oxidant effects (Long et al., 2019). Although the direct effect on IBS has not been known, it is known to have significant efficacy as a supplementary medicine in the treatment of GI protection and IBD (Wang et al., 2022). It also inhibits the TRPV1 channel (Wang et al., 2020). Ginsenoside Rb1 are representative active ingredients of *Panax ginseng* C.A.Mey. and have known to have good therapeutic effects on cardiovascular disease, cancer, glaucoma, and stroke (Ahmed et al., 2016). It has known to have therapeutic effects in IBD, although its efficacy against IBS has unknown (Toyokawa et al., 2019). In addition, it inhibits TRPV1 channels (Huang et al., 2010). In summary, berberine has been known to be effective in IBS, and the standard compounds related to TRPA1 and NaV are 6-gingerol, liquiritin, Isoliquiritigenin and berberine. In the future, the IBS efficacy and the efficacy in IBS-related

ion channels of each of these standard compounds will be systematically confirmed.

Pinellia ternata (Thunb.) Makino is an effective herbal medicine for removing dampness-phlegm that is often used in traditional medicine (Kim et al., 2006). It has the effects of insomnia, anti-tumor, antibacterial, anti-inflammatory, and sedative, and is also used to treat cough and emotional diseases (Kim et al., 2006; Lin et al., 2021; Choi et al., 2022). It is known to be effective in treating vomiting and GI motility diseases in the GI tract (Choi et al., 2022). There are no research results related to irritable bowel syndrome by *Pinellia ternata* (Thunb.) Makino itself, and studies related to IBS have been conducted through a systems pharmacology strategy by BHSST (Li et al., 2019). The ion channel effect by *Pinellia ternata* (Thunb.) Makino is known to increase Ca²⁺ in cells by increasing the influx of Ca²⁺ into cells, and thus, it has been reported to increase voltage dependent Ca²⁺ and Ca²⁺ activated K⁺ currents (Shi et al., 1994). *Scutellaria baicalensis* Georgi is a plant of the genus Lamiales, and roots are mainly used as medicine (Zhao et al., 2019). It is still important as a traditional medicine that has the function of removing heat and poison. It is composed of more than 110 flavonoids, including baicalin, baicalein, wogonoside, wogonin, etc (Song et al., 2020). It is mainly used as a treatment for hepatitis, high blood pressure, acute respiratory disease, acute GI inflammation, diarrhea, and vomiting, and has recently been widely used in the form of tea that removes heat and poison, suppresses inflammation, and promotes digestion (Zhao et al., 2019). It is also used for the prevention and treatment of IBS-related intestinal and nervous diseases due to the functions of nerve protection and microbial control (Liang et al., 2019), and is also widely used for the treatment of diarrhea by IBS along with *Coptis chinensis* Franch (Chen et al., 2015). Components of *Scutellaria baicalensis* Georgi are known to regulate BKCa and ATP sensitive K⁺ ion pathways (Ding et al., 2019; C.Y. Wu et al., 2020), but there are not many studies on ion channel efficacy. *Zingiber officinale* Roscoe is a widely used medicinal plant and a powerful immune-enhancing spice. It has been used for centuries to treat nausea and has a variety of biological activities, including antioxidant, anti-inflammatory, antimicrobial, anticancer, neuroprotective, cardiovascular protective, respiratory protective, anti-obesity, antidiabetic, anti-nausea, and antiemetic activities (Mao et al., 2019). It can be used fresh or dried, and dried ginger contains 5.2 times more phenolic compounds that are closely related to polyphenol content than fresh ginger, making it even more effective in antioxidant activity (Mao et al., 2019). Additionally, it and its compound, 6-shogaol, are known to improve GI function by reducing inflammatory responses in the GI tract (Huh et al., 2020). It has also been shown to have a beneficial effect on IBS (M. Zhang et al., 2020). It regulates TRPV1 and TRPM8 ion channels that are closely related to temperature and pain sensation, which can improve pain (Liu et al., 2022). *Panax ginseng* C.A. Mey. has been known to have the effects of increasing energy and vitality when tired and stressed, that is, improving the overall quality of life (Yennurajalingam et al., 2017; Arring et al., 2018). In some past papers, it affects the cardiovascular, peripheral and central nervous systems, and the immune system, increasing various biological activities (Chen, 1996; Gillis, 1997). It is also reported to affect the GI system (Kim et al., 2007). In GI systems, it also has a therapeutic effect on the management of IBS (Yu et al., 2021). Ginsenoside Rg3, one of the components of *Panax ginseng* C.A.Mey., controls various voltage-gated and ligand-gated ion channels by combining with ion channel blocker binding sites, toxin-binding sites, and channel gating regions (Nah, 2014). *Glycyrrhiza uralensis* Fisch. ex DC. is a herbal medicine commonly used to maintain health and vitality regardless of age (Jiang et al., 2021). Past studies have shown that it has the effects of antioxidants, anti-inflammatory, antiviral, and anti-diabetes, as well as liver protection and skin whitening. It also has culinary properties as a spice (Gou et al., 2021; J. Wu et al., 2020). It contains many flavonoids such as liquiritigenin, isoliquiritigenin, liquiritin, isoliquiritin, and glycyrrhizin, which are important active components (Yue et al., 2021). *Glycyrrhiza uralensis* Fisch. ex DC. has few studies on its efficacy in IBS, but it

is known to have the function of relieving colitis with antioxidant and anti-inflammatory effects by controlling intestinal microorganisms, NF- κ B, and FXR/P-gp mechanisms (Liu et al., 2022; Shi et al., 2022). According to the role of ion channels, it has the effect of inhibiting voltage-gated Na⁺ 1.4 and voltage-gated K⁺ 1.3 ion channels (Fu et al., 2013; Zhu et al., 2018). *Ziziphus jujuba* Mill. is a highly nutritious and functional fruit that is widely cultivated, particularly in China. It is known to improve sleep quality, eliminate toxins, and beautify the skin (Lu et al., 2021a,b). According to recent research, one of the main biologically active components in *Ziziphus jujuba* Mill. fruit, polysaccharides, exhibits various biological effects, including immune regulation, antioxidant, anti-tumor, liver protection, blood sugar-lowering, and gastrointestinal protection effects (Ji et al., 2017). In addition, the polysaccharides have been found to improve trinitrobenzene sulfonic acid (TNBS)-induced colitis (Yue et al., 2015). Furthermore, network pharmacology approaches have indicated that its components may have an appetite-regulating effect through the regulation of TRP channels, which are inflammatory mediators, as revealed by KEGG analysis (Zhu et al., 2022). *Coptis chinensis* Franch, a representative medicinal herb known in traditional Korean medicine for its ability to lower fever and remove toxicity (Qi et al., 2022), has been reported to have anti-diabetic, relaxant, pyretic, antibacterial, and antiviral effects (Friedemann et al., 2014). It can effectively regulate not only gut bacteria but also the microenvironment of the gut, such as Peyer's patches and gut epithelial-related immune responses (Chen et al., 2021). However, it has not been experimentally clarified whether *Coptis chinensis* Franch is effective for IBS. Berberine extracted from *Coptis chinensis* Franch not only inhibits Ca²⁺ channels and extracellular Ca²⁺ influx but also reduces delayed depolarization caused by calcium overload (He et al., 2022). BHSST is a traditional medical prescription made up of eight herbal medicines, and the efficacy of BHSST may be manifested by the synergy of the combination of these eight herbal medicines. Some of the eight components affect the GI tract, and some of them are also effective in IBS. Looking at the efficacy of ion channels, *Zingiber officinale* Roscoe (fresh or dried), *Glycyrrhiza uralensis* Fisch. ex DC., and *Ziziphus jujuba* Mill. are known to control TRP and NaV ion channels, which are mentioned in this study. Therefore, it is assumed that these three components may be probably involved in the reduction of IBS induced pain by BHSST. However, in order to confirm the exact involvement of any components, we will check the efficacy of each components for TRPV1, V4, A1 and NaV1.5 and 1.7 in the future, and then we will be able to know which components of BHSST affects IBS.

In conclusion, this study found that BHSST increased colon length and normalized colon weight in a mouse model of irritable bowel syndrome (IBS) with diarrhea as the main symptom. Stool scores recovered, food intake remained unchanged, and the body weight of the mice increased. Inflammation-related levels also returned to normal, and pain-related behaviors were reduced. TRPA1, NaV1.5, and NaV1.7 ion channels may be involved in controlling visceral hypersensitivity. IBS is a heterogeneous group of diseases with various symptoms and pathophysiologies. Thus, combining drugs for the related pathophysiology or symptoms can be effective in treating IBS. Traditional medicines such as BHSST may be safe and effective in treating abdominal pain and discomfort caused by GI diseases such as IBS.

CRedit authorship contribution statement

Na Ri Choi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Min Ji Kwon:** Methodology, Software, Validation, Investigation, Data curation. **Woo-Gyun Choi:** Writing – review & editing, Visualization. **Sang Chan Kim:** Methodology, Software, Investigation, Writing – review & editing. **Jae-Woo Park:** Investigation, Resources, Data curation, Writing – review & editing. **Joo Hyun Nam:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Byung Joo Kim:**

Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

Data will be made available on request.

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References

- Ahmed, T., Raza, S.H., Maryam, A., Setzer, W.N., Braid, N., Nabavi, S.F., de Oliveira, M. R., Nabavi, S.M., 2016. Ginsenoside Rb1 as a neuroprotective agent: a review. *Brain Res. Bull.* 125, 30–43.
- Ai, X., Yu, P., Peng, L., Luo, L., Liu, J., Li, S., Lai, X., Luan, F., Meng, X., 2021. Berberine: a review of its pharmacokinetics properties and therapeutic potentials in diverse vascular diseases. *Front. Pharmacol.* 12, 762654.
- Akash, M.S.H., Fiayyaz, F., Rehman, K., Sabir, S., Rasool, M.H., 2019. Gut microbiota and metabolic disorders: advances in therapeutic interventions. *Crit. Rev. Immunol.* 39 (4), 223–237.
- Aoyama, T., Nishikawa, K., Takiguchi, N., Tanabe, K., Imano, M., Fukushima, R., Sakamoto, J., Oba, M.S., Morita, S., Kono, T., Tsuburaya, A., 2014. Double-blind, placebo-controlled, randomized phase II study of TJ-14 (hangeshashinto) for gastric cancer chemotherapy-induced oral mucositis. *Cancer Chemother. Pharmacol.* 73 (5), 1047–1054.
- Arring, N.M., Millstine, D., Marks, L.A., Nail, L.M., 2018. Ginseng as a treatment for fatigue: a systematic review. *J. Alternative Compl. Med.* 24 (7), 624–633.
- Beckers, A.B., Weerts, Z.Z.R.M., Helyes, Z., Masclee, A.A.M., Keszthelyi, D., 2017. Review article: transient receptor potential channels as possible therapeutic targets in irritable bowel syndrome. *Aliment. Pharmacol. Ther.* 46 (10), 938–952.
- Canavan, C., West, J., Card, T., 2014. The epidemiology of irritable bowel syndrome. *Clin. Epidemiol.* 6, 71–80.
- Cenac, N., Bautzova, T., Le Faouder, P., Veldhuis, N.A., Poole, D.P., Rolland, C., Bertrand, J., Liedtke, W., Dubourdeau, M., Bertrand-Michel, J., Zecchi, L., Stanghellini, V., Bunnett, N.W., Barbara, G., Vergnolle, N., 2015. Quantification and potential functions of endogenous agonists of transient receptor potential channels in patients with irritable bowel syndrome. *Gastroenterology* 149 (2), 433–444.
- Chaudhury, D., Liu, H., Han, M.H., 2015. Neuronal correlates of depression. *Cell. Mol. Life Sci.* 72 (24), 4825–4848.
- Chen, C., Lu, M., Pan, Q., Fichna, J., Zheng, L., Wang, K., 2015. Berberine improves intestinal motility and visceral pain in the mouse models mimicking diarrhea predominant Irritable Bowel Syndrome (IBS-D) symptoms in an opioid-receptor dependent manner. *PLoS One* 10, e0145556.
- Chen, F.P., Chen, F.J., Jong, M.S., Tsai, H.L., Wang, J.R., Hwang, S.J., 2009. Modern use of Chinese herbal formulae from Shang-Han Lun. *Chin. Med. J.* 122 (16), 1889–1894.
- Chen, G., Bei, B., Feng, Y., Li, X., Jiang, Z., Si, J.Y., Qing, D.G., Zhang, J., Li, N., 2019. Glycyrrhetic acid maintains intestinal homeostasis via HuR. *Front. Pharmacol.* 10, 535.
- Chen, G., Zhu, L., Liu, Y., Zhou, Q., Chen, H., Yang, J., 2009. Isoliquiritigenin, a flavonoid from licorice, plays a dual role in regulating gastrointestinal motility in vitro and in vivo. *Phytother. Res.* 23 (4), 498–506.
- Chen, K., Yang, R., Shen, F.Q., Zhu, H.L., 2020. Advances in pharmacological activities and mechanisms of glycyrrhizic acid. *Curr. Med. Chem.* 27 (36), 6219–6243.
- Chen, M., Ruan, G., Chen, L., Ying, S., Li, G., Xu, F., Xiao, Z., Tian, Y., Lv, L., Ping, Y., Cheng, Y., Wei, Y., 2022. Neurotransmitter and intestinal interactions: focus on the microbiota-gut-brain Axis in irritable bowel syndrome. *Front. Endocrinol.* 13, 817100.
- Chen, Q., Ren, R., Zhang, Q., Wu, J., Zhang, Y., Xue, M., Yin, D., Yang, Y., 2021. *Coptis chinensis* Franch polysaccharides provide a dynamically regulation on intestinal microenvironment, based on the intestinal flora and mucosal immunity. *J. Ethnopharmacol.* 267, 113542.
- Chen, X., 1996. Cardiovascular protection by ginsenosides and their nitric oxide releasing action. *Clin. Exp. Pharmacol. Physiol.* 23 (8), 728–732.
- Choi, N.R., Park, J., Ko, S.J., Kim, J.N., Choi, W., Park, J.W., Kim, B.J., 2022. Prediction of the medicinal mechanisms of *Pinellia ternata* Breitenbach, a traditional medicine for gastrointestinal motility disorders, through network pharmacology. *Plants (Basel)* 11 (10), 1348.
- Choi, Y.H., Bae, J.K., Chae, H.S., Choi, Y.O., Nhoek, P., Choi, J.S., Chin, Y.W., 2016. Isoliquiritigenin ameliorates dextran sulfate sodium-induced colitis through the inhibition of MAPK pathway. *Int. Immunopharm.* 31, 223–232.

- Chun, E., Yoon, S., Parveen, A., Jin, M., 2018. Alleviation of irritable bowel syndrome-like symptoms and control of gut and brain responses with oral administration of *Dolichos lablab* L. In a mouse model. *Nutrients* 10 (10), 1475.
- Coutinho, S.V., Meller, S.T., Gebhart, G.F., 1996. Intracolonic zymosan produces visceral hyperalgesia in the rat that is mediated by spinal NMDA and non-NMDA receptors. *Brain Res.* 736 (1–2), 7–15.
- Ding, L.Q., Jia, C.L., Zhang, Y., Wang, W.J., Zhu, W.L., Chen, Y., Zhang, T., 2019. Baicalin relaxes vascular smooth muscle and lowers blood pressure in spontaneously hypertensive rats. *Biomed. Pharmacother.* 111, 325–330.
- Drossman, D.A., Hasler, W.L., 2016. Rome IV-functional GI disorders: disorders of gut-brain interaction. *Gastroenterology* 150 (6), 1257–1261.
- Durojaye, O.A., Nwaguma, B.C., Joshua, P.E., Ogidigo, J.O., Cosmas, S., Asomadu, R.O., Obeta, J.N., 2019. Evaluation of 6-Gingerol and its modified analogues as therapeutic candidates against *Schistosoma mansoni* phosphofructokinase. *VaccinMonitor* 28 (1), 38–47.
- Emery, L.Q., Jia, C.L., Wood, J.N., 2016. Nav1.7 and other voltage-gated sodium channels as drug targets for pain relief. *Expert Opin. Ther. Targets* 20 (8), 975–983.
- Flayyaz, F., Rehman, K., Sharif, H., Irshad, K., Kamal, S., Akash, M.S.H., 2021. Endocrine Disrupting Chemicals-Induced Metabolic Disorders and Treatment Strategies. Springer Cham, Switzerland, pp. 125–133.
- Friedemann, T., Otto, B., Klätschke, K., Schumacher, U., Tao, Y., Leung, A.K.M., Obeth, T., Schröder, S., 2014. *Coptis chinensis* Franch. exhibits neuroprotective properties against oxidative stress in human neuroblastoma cells. *J. Ethnopharmacol.* 155 (1), 607–615.
- Fu, X.X., Du, L.L., Zhao, N., Dong, Q., Liao, Y.H., Du, Y.M., 2013. 18 β -Glycyrrhetic acid potentially inhibits Kv1.3 potassium channels and T cell activation in human Jurkat T cells. *J. Ethnopharmacol.* 148 (2), 647–654.
- Fuentes, I.M., Christianson, J.A., 2016. Ion channels, ion channel receptors, and visceral hypersensitivity in irritable bowel syndrome. *Neuro Gastroenterol. Motil.* 28 (11), 1613–1618.
- Fukamachi, H., Matsumoto, C., Omiya, Y., Arimoto, T., Morisaki, H., Kataoka, H., Kadena, M., Funatsu, T., Fukutake, M., Kase, Y., Kuwata, H., 2015. Effects of hangeshashinto on growth of oral microorganisms. *Evid. Based Compl. Altern. Med.* 2015, 512947.
- Gillis, C.N., 1997. Panax ginseng pharmacology: a nitric oxide link? *Biochem. Pharmacol.* 54 (1), 1–8.
- Gou, S.H., He, M., Li, B.B., Zhu, N.Y., Ni, J.M., 2021. Hepatoprotective effect of total flavonoids from *Glycyrrhiza uralensis* Fisch in liver injury mice. *Nat. Prod. Res.* 35 (24), 6083–6087.
- Gunthorpe, M.J., Benham, C.D., Randall, A., Davis, J.B., 2002. The diversity in the vanilloid (TRPV) receptor family of ion channels. *Trends Pharmacol. Sci.* 23 (4), 183–191.
- Gyamfi, O.A., Bortey-Sam, N., Donkor, A.B., White, C.W., Logue, B.A., 2020. Analysis of TRPA1 antagonist, A-967079, in plasma using high-performance liquid chromatography tandem mass-spectrometry. *J. Pharm. Anal.* 10 (2), 157–163.
- He, J., Li, S., Ding, Y., Tong, Y., Li, X., 2022. Research progress on natural products' therapeutic effects on atrial fibrillation by regulating ion channels. *Cardiovasc. Ther.* 2022, 4559809.
- Heber, S., Ciotu, C.L., Hartner, G., Gold-Binder, M., Ninidze, N., Gleiss, A., Kress, H.G., Fischer, M.J.M., 2020. TRPV1 antagonist BCTC inhibits pH 6.0-induced pain in human skin. *Pain* 161 (7), 1532–1541.
- Hellstrom, P.M., 2019. Pathophysiology of the irritable bowel syndrome-reflections of today. *Bailliere's best Pract. Res. Clin. Gastroenterol.* 40–41, 101620.
- Hitomi, S., Ono, K., Terawaki, K., Matsumoto, C., Mizuno, K., Yamaguchi, K., Imai, R., Omiya, Y., Hattori, T., Kase, Y., Inenaga, K., 2017. [6]-gingerol and [6]-shogaol, active ingredients of the traditional Japanese medicine hangeshashinto, relief oral ulcerative mucositis-induced pain via action on Na⁺ channels. *Pharmacol. Res.* 117, 288–302.
- Hitomi, S., Ono, K., Yamaguchi, K., Terawaki, K., Imai, R., Kubota, K., Omiya, Y., Hattori, T., Kase, Y., Inenaga, K., 2016. The traditional Japanese medicine hangeshashinto alleviates oral ulcer-induced pain in a rat model. *Arch. Oral Biol.* 66, 30–37.
- Huan, P., Wang, L., He, Z., He, J., 2022. The role of gut microbiota in the progression of Parkinson's disease and the mechanism of intervention by traditional Chinese medicine. *Neuropsychiatric Dis. Treat.* 18, 1507–1520.
- Huang, J., Qiu, L., Ding, L., Wang, S., Wang, J., Zhu, Q., Song, F., Hu, J., 2010. Ginsenoside Rb1 and paeoniflorin inhibit transient receptor potential vanilloid-1-activated IL-8 and PGE₂ production in a human keratinocyte cell line HaCaT. *Int. Immunopharm.* 10 (10), 1279–1283.
- Hugon, P., Dufour, J.C., Colson, P., Fournier, P.E., Sallah, K., Raoult, D., 2015. A comprehensive repertoire of prokaryotic species identified in human beings. *Lancet Infect. Dis.* 15 (10), 1211–1219.
- Huh, E., Choi, J.G., Noh, D., Yoo, H.S., Ryu, J., Kim, N.J., Kim, H., Oh, M.S., 2020. Ginger and 6-shogaol protect intestinal tight junction and enteric dopaminergic neurons against 1-methyl-4-phenyl 1,2,3,6-tetrahydropyridine in mice. *Nutr. Neurosci.* 23 (6), 455–464.
- Jahng, J., Jung, I.S., Choi, E.J., Conklin, J.L., Park, H., 2012. The effects of methane and hydrogen gases produced by enteric bacteria on ileal motility and colonic transit time. *Neuro Gastroenterol. Motil.* 24 (2), e92.
- Jang, D.E., Bae, J.H., Chang, Y.J., Lee, Y.H., Nam, K.T., Kim, I.Y., Seong, J.K., Lee, Y.C., Yeom, S.C., 2018. Neuronal nitric oxide synthase is a novel biomarker for the interstitial cells of Cajal in stress-induced diarrhea-dominant irritable bowel syndrome. *Dig. Dis. Sci.* 63 (3), 619–627.
- Jeon, Y.J., Lee, J.S., Cho, Y.R., Lee, S.B., Kim, W.Y., Roh, S.S., Joung, J.Y., Lee, H.D., Moon, S.O., Cho, J.H., Son, C.G., 2019. Banha-sasim-tang improves gastrointestinal function in loperamide-induced functional dyspepsia mouse model. *J. Ethnopharmacol.* 28 (238), 111834.
- Ji, X., Peng, Q., Yuan, Y., Shen, J., Xie, X., Wang, M., 2017. Isolation, structures and bioactivities of the polysaccharides from jujube fruit (*Ziziphus jujuba* Mill.): a review. *Food Chem.* 227, 349–357.
- Jiang, L., Akram, W., Luo, B., Hu, S., Faruque, M.O., Ahmad, S., Yasin, N.A., Khan, W.U., Ahmad, A., Shikov, A.N., Chen, J., Hu, X., 2021. Metabolomic and pharmacologic insights of aerial and underground parts of *Glycyrrhiza uralensis* Fisch. ex DC. for maximum utilization of medicinal Resources. *Front. Pharmacol.* 12, 658670.
- Jin, M., Wu, Z., Chen, L., Jaimes, J., Collins, D., Edgar, T., Walters, E.T., Roger, G., O'Neil, R.G., 2011. Determinants of TRPV4 activity following selective activation by small molecule agonist GSK1016790A. *PLoS One* 6 (2), e16713.
- Kase, Y., Hayakawa, T., Takeda, S., Ishige, A., Aburada, M., Okada, M., 1996. Pharmacological studies on antidiarrheal effects of Hange-shashin-to. *Biol. Pharm. Bull.* 19 (10), 1367–1370.
- Kase, Y., Saitoh, K., Ishige, A., Komatsu, Y., 1998. Mechanisms by which Hange-shashin-to reduces prostaglandin E2 levels. *Biol. Pharm. Bull.* 21 (12), 1277–1281.
- Kamide, D., Yamashita, T., Araki, K., Tomifuji, M., Shiotani, A., 2017. Hangeshashinto (TJ-14) prevents radiation-induced mucositis by suppressing cyclooxygenase-2 expression and chemotaxis of inflammatory cells. *Clin. Transl. Oncol.* 19 (11), 1329–1336.
- Kawashima, K., Fujimura, Y., Makino, T., Kano, Y., 2006. Pharmacological properties of traditional medicine (XXXII): protective effects of hangeshashinto and the combinations of its major constituents on gastric lesions in rats. *Biol. Pharm. Bull.* 29 (9), 1973–1975.
- Kawashima, K., Nomura, A., Makino, T., Saito, K.I., Kano, Y., 2004. Pharmacological properties of traditional medicine (XXIX): effect of hange-shashin-to and the combinations of its herbal constituents on rat experimental colitis. *Biol. Pharm. Bull.* 27 (10), 1599–1603.
- Kim, H.S., Parajuli, S.P., Yeum, C.H., Park, J.S., Jeong, H.S., So, I., Kim, K.W., Jun, J.Y., Choi, S., 2007. Effects of ginseng total saponins on pacemaker currents of interstitial cells of Cajal from the small intestine of mice. *Biol. Pharm. Bull.* 30 (11), 2037–2042.
- Kim, J.N., Nam, J.H., Lee, J.R., Kim, S.C., Kim, B.J., 2020. The traditional medicine Banhasasim-Tang depolarizes pacemaker potentials of cultured interstitial cells of Cajal through M3 muscarinic and 5-HT3 receptors in murine small intestine. *Digestion* 101 (5), 536–551.
- Kim, S.K., Joung, J.Y., Ahn, Y.C., Jung, I.C., Son, C.G., 2021. Beneficial potential of *Banha-Sasim-tang* for stress-sensitive functional dyspepsia via modulation of Ghrelin: a randomized controlled trial. *Front. Pharmacol.* 12, 636752.
- Kim, Y.J., Shin, Y.O., Ha, Y.W., Lee, S., Oh, J.K., Kim, Y.S., 2006. Anti-obesity effect of *Pinellia ternata* extract in Zucker rats. *Biol. Pharm. Bull.* 29 (6), 1278–1281.
- Kono, T., Kaneko, A., Matsumoto, C., Miyagi, C., Ohbuchi, K., Mizuhara, Y., Miyano, K., Uezono, Y., 2014. Multitargeted effects of hangeshashinto for treatment of chemotherapy-induced oral mucositis on inducible prostaglandin E2 production in human oral keratinocytes. *Integr. Cancer Ther.* 13 (5), 435–445.
- Laird, J.M.A., Martinez-Caro, L., Garcia-Nicas, E., Cervero, F., 2001. A new model of visceral pain and referred hyperalgesia in the mouse. *Pain* 92 (3), 335–342.
- Lee, K.G., Cui, X., Lim, J.P., 2002. Effect of the concurrent administration of Banhasasim-tang with cimetidine on gastric ulcer in rats. *Kor. J. Ori. Med. Physiol. Pathol.* 16 (3), 572–576.
- Lee, K.N., Lee, O.Y., 2014. Intestinal microbiota in pathophysiology and management of irritable bowel syndrome. *World J. Gastroenterol.* 20 (27), 8886–8897.
- Li, B., Rui, J., Ding, X., Yang, X., 2019. Exploring the multicomponent synergy mechanism of Banxia Xiexin Decoction on irritable bowel syndrome by a systems pharmacology strategy. *J. Ethnopharmacol.* 233, 158–168.
- Li, H., Xia, J., Zhang, W., Gu, L., Pan, C., Tian, Y., 2017. Xifeng Huashi Formula presents therapeutic effect on diarrhea predominant irritable bowel syndrome through regulating brain-gut axis and SCF/C-KIT signaling pathway. *Int. J. Clin. Exp. Med.* 10 (11), 15789–15797.
- Liang, S., Deng, X., Lei, L., Zheng, Y., Ai, J., Chen, L., Xiong, H., Mei, Z., Cheng, Y.C., Ren, Y., 2019. The comparative study of the therapeutic effects and mechanism of baicalin, baicalein, and their combination on ulcerative colitis rat. *Front. Pharmacol.* 10, 1466–1481.
- Lin, Y.H., Chen, C., Zhao, X., Mao, Y.F., Xiang, G.X., Yang, M.Q., Song, Y.M., 2021. Efficacy and safety of Banxia formulae for insomnia: a systematic review and meta-analysis of high-quality randomized controlled trials. *Evid. Based Compl. Alternat. Med.* 2021, 8833168.
- Liu, X., Meng, X., Su, X., Ren, K., Ning, C., Qi, X., Zhang, S., 2022. The mechanism of ginger and its processed products in the treatment of estradiol valerate coupled with oxytocin-induced dysmenorrhea in mice via regulating the TRP ion channel-mediated ERK1/2/NF- κ B signaling pathway. *Food Funct.* 13 (21), 11236–11248.
- Liu, Z., Wang, P., Lu, S., Guo, R., Gao, W., Tong, H., Yin, Y., Han, X., Liu, T., Chen, X., Zhu, M.X., Yang, Z., 2020. Liquiritin, a novel inhibitor of TRPV1 and TRPA1, protects against LPS-induced acute lung injury. *Cell Calcium* 88, 102198.
- Long, Y., Xiang, Y., Tan, Y.J., Li, N., Yang, Q.Y., 2019. Research progress on pharmacological action and new dosage forms of baicalin. *Chin. Tradit. Herb. Drugs* 50, 6142–6148.
- Lu, Y., Bao, T., Mo, J., Ni, J., Chen, W., 2021. Research advances in bioactive components and health benefits of jujube (*Ziziphus jujuba* Mill.) fruit. *J. Zhejiang Univ. - Sci. B* 22 (6), 431–449.
- Lu, Q., Wu, X., Han, W., Zhang, W., Wang, Y., Kong, D., Fan, Z., 2021. Effect of *Glycyrrhiza uralensis* against ulcerative colitis through regulating the signaling pathway of FXR/P-gp. *Am. J. Transl. Res.* 13 (8), 9296–9305.
- Mao, Q.Q., Xu, X.Y., Cao, S.Y., Gan, R.Y., Corke, H., Beta, T., Li, H.B., 2019. Bioactive compounds and bioactivities of ginger (*Zingiber officinale* Roscoe). *Foods* 8 (6), 185.

- Marchesi, J.R., 2010. Prokaryotic and eukaryotic diversity of the human gut. *Adv. Appl. Microbiol.* 72, 43–62.
- Meng, J., Qiu, S., Zhang, L., You, M., Xing, H., Zhu, J., 2022. Berberine alleviate Cisplatin-induced peripheral neuropathy by modulating inflammation signal via TRPV1. *Front. Pharmacol.* 12, 774795.
- Miyamura, Y., Hitomi, S., Omiya, Y., Ujihara, I., Kokabu, S., Morimoto, Y., Ono, K., 2021. Isoliquiritigenin, an active ingredient of Glycyrrhiza, elicits antinociceptive effects via inhibition of NaV channels. *Naunyn-Schmiedeberg's Arch. Pharmacol.* 394 (5), 967–980.
- Mohammadzadeh, N., Kalani, B.S., Bolori, S., Azadegan, A., Gholami, A., Mohammadzadeh, R., Jazi, F.M., 2019. Identification of an intestinal microbiota signature associated with hospitalized patients with diarrhea. *Acta Microbiol. Immunol. Hung.* 66 (2), 189–202.
- Morera, E., De Petrocellis, L., Morera, L., Moriello, A.S., Nalli, M., Di Marzo, V., Ortá, G., 2012. Synthesis and biological evaluation of [6]-gingerol analogues as transient receptor potential channel TRPV1 and TRPA1 modulators. *Bioorg. Med. Chem. Lett.* 22 (4), 1674–1677.
- Morgan, V., Pickens, D., Gautam, S., Kessler, R., Mertz, H., 2005. Amitriptyline reduces rectal pain related activation of the anterior cingulate cortex in patients with irritable bowel syndrome. *Gut* 54 (5), 601–607.
- Nah, S.Y., 2014. Ginseng ginsenoside pharmacology in the nervous system: involvement in the regulation of ion channels and receptors. *Front. Physiol.* 5, 98.
- Nash, D.T., Nash, S.D., 2008. Ranolazine for chronic stable angina. *Lancet* 372 (9646), 1335–1341.
- Park, B.K., Chun, E., Choi, J.J., Shin, Y., Kho, Y.T., Oh, S.H., Kim, S.Y., Lee, T.H., Kim, T.W., Shin, E., Do, S.G., Jin, M., 2017. Administration of *Wasabia koreana* ameliorates irritable bowel syndrome-like symptoms in a Zymosan-induced mouse model. *J. Med. Food* 20 (5), 474–484.
- Park, J.W., Ko, S.J., Han, G., Yeo, I., Ryu, B., Kim, J., 2013. The effects of *Banha-sasim-tang* on dyspeptic symptoms and gastric motility in cases of functional dyspepsia: a randomized, double-blind, placebo-controlled, and two-center trial. *Evid. Based Compl. Alternat. Med.* 2013, 265035.
- Park, J.W., Ryu, B., Yeo, I., Jerng, U.M., Han, G., Oh, S., Lee, J., Kim, J., 2010. *Banha-sasim-tang* as an herbal formula for the treatment of functional dyspepsia: a randomized, double-blind, placebo-controlled, two-center trial. *Trials* 11, 83.
- Peng, F., Du, Q., Peng, C., Wang, N., Tang, H., Xie, X., Shen, J., Chen, J., 2015. A review: the pharmacology of isoliquiritigenin. *Phytother. Res.* 29 (7), 969–977.
- Pozuelo, M., Panda, S., Santiago, A., Mendez, S., Accarino, A., Santos, J., Guarner, F., Azpiroz, F., Manichanh, C., 2015. Reduction of butyrate- and methane-producing microorganisms in patients with Irritable Bowel Syndrome. *Sci. Rep.* 5, 12693.
- Promdam, N., Panichayupakaranant, P., 2022. [6]-Gingerol: a narrative review of its beneficial effect on human health. *Food Chem. Adv.* 1, 100043.
- Pruzanski, W., Stefanski, E., Vadas, P., Ramamurthy, N.S., 1997. Inhibition of extracellular release of proinflammatory secretory phospholipase A2 (sPLA2) by sulfasalazine: a novel mechanism of anti-inflammatory activity. *Biochem. Pharmacol.* 53 (12), 1901–1907.
- Qi, Y., Ni, S., Heng, X., Qu, S., Ge, P., Zhao, X., Yao, Z., Guo, R., Yang, N., Zhang, Q., Zhu, H., 2022. Uncovering the potential mechanisms of *Coptis chinensis* Franch. for serious mental illness by network pharmacology and pharmacology-based analysis. *Drug Des. Dev. Ther.* 16, 325–342.
- Qin, J., Chen, J., Peng, F., Sun, C., Lei, Y., Chen, G., Li, G., Yin, Y., Lin, Z., Wu, L., Li, J., Liu, W., Peng, C., Xie, X., 2022. Pharmacological activities and pharmacokinetics of liquiritin: a review. *J. Ethnopharmacol.* 293, 115257.
- Ramalingam, M., Kim, H., Lee, Y., Lee, Y.I., 2018. Phytochemical and pharmacological role of liquiritigenin and isoliquiritigenin from *Radix Glycyrrhizae* in human health and disease models. *Front. Aging Neurosci.* 10, 348.
- Rothschild, D., Weissbrod, O., Barkan, E., Kurilshikov, A., Korem, T., Zeevi, D., Costea, P. I., Godneva, A., Kalka, I.N., Bar, N., Shilo, S., Lador, D., Vila, A.V., Zmora, N., Pevsner-Fischer, M., Israeli, D., Kosover, N., Malka, G., Wolf, B.C., Avnit-Sagi, T., Lotan-Pompan, M., Weinberger, A., Halpern, Z., Carmi, S., Fu, J., Wijmenga, C., Zhernakova, A., Elinav, E., Segal, E., 2018. Environment dominates over host genetics in shaping human gut microbiota. *Nature* 555 (7695), 210–215.
- Sandor, Z., Dekany, A., Kelemen, D., Bencsik, T., Papp, R., Bartho, L., 2016. The TRPA1 activator allyl isothiocyanate (AITC) contracts human Jejunal muscle: pharmacological analysis. *Basic Clin. Pharmacol. Toxicol.* 119 (3), 341–342.
- Schmulson, M.J., Drossman, D.A., 2017. What is new in Rome IV. *J. Neurogastroenterol. Motil.* 23 (2), 151–163.
- Sheng, Y., Wu, T., Dai, Y., Xu, L., Zhong, Y., Xue, Y., Tian, Y., 2020. 6-Gingerol alleviates inflammatory injury in DSS-induced ulcerative colitis mice by regulating NF- κ B signaling. *Ann. Palliat. Med.* 9 (4), 1944–1952.
- Shi, G., Kong, J., Wang, Y., Xuan, Z., Xu, F., 2022. Glycyrrhiza uralensis Fisch. alleviates dextran sulfate sodium-induced colitis in mice through inhibiting of NF- κ B signaling pathways and modulating intestinal microbiota. *J. Ethnopharmacol.* 298, 115640.
- Shin, M.R., Seo, B.I., Son, C.G., Roh, S.S., An, H.J., 2017. *Banha-sasim-tang* treatment reduces the severity of esophageal mucosal ulcer on chronic acid reflux esophagitis in rats. *BioMed Res. Int.* 2017, 7157212.
- Shi, Y.L., Xu, Y.F., Zhang, H., 1994. Effect of *Pinellia ternata* lectin on membrane currents of mouse motor nerve terminals. *Sci. China E B.* 37 (4), 448–453.
- Shih, Y.C., Barghout, V.E., Sandler, R.S., Jhingran, P., Sasane, M., Cook, S., Gibbons, D. C., Halpern, M., 2002. Resource utilization associated with irritable bowel syndrome in the United States 1987–1997. *Dig. Dis. Sci.* 47 (8), 1705–1715.
- Song, J.W., Long, J.Y., Xie, L., Zhang, L.L., Xie, Q.X., Chen, H.J., Deng, M., Li, X.F., 2020. Applications, phytochemistry, pharmacological effects, pharmacokinetics, toxicity of *Scutellaria baicalensis* Georgi. and its probably potential therapeutic effects on COVID-19: a review. *Chin. Med.* 15, 102.
- Strege, P.R., Ou, Y., Sha, L., Rich, A., Gibbons, S.J., Szurszewski, J.H., Sarr, M.G., Farrugia, G., 2003. Sodium current in human intestinal interstitial cells of Cajal. *Am. J. Physiol. Gastrointest. Liver Physiol.* 285 (6), G1111–G1121.
- Stucky, C.L., Dubin, A.E., Jeske, N.A., Malin, S.A., McKemy, D.D., Story, G.M., 2009. Roles of transient receptor potential channels in pain. *Brain Res. Rev.* 60 (1), 2–23.
- Sung, J.J.Y., Leung, W.K., Ching, J.Y.L., Lao, L., Zhang, G., Wu, J.C.Y., Liang, S.M., Xie, H., Ho, Y.P., Chan, L.S., Berman, B., Chan, F.K.L., 2004. Agreements among traditional Chinese medicine practitioners in the diagnosis and treatment of irritable bowel syndrome. *Aliment. Pharmacol. Ther.* 20 (10), 1205–1210.
- Takasuna, K., Kasai, Y., Kitano, Y., Mori, K., Kobayashi, R., Hagiwara, T., Kakhata, K., Hirohashi, M., Nomura, M., Nagai, E., Kamataki, T., 1995. Protective effects of kampo medicines and baicalin against intestinal toxicity of a new anticancer camptothecin derivative, irinotecan hydrochloride (CPT-11), in rats. *Jpn. J. Cancer Res.* 86 (10), 978–984.
- Tang, Z.P., 2009. Traditional Chinese medicine clinical experience of the treatment for irritable bowel syndrome. *Chin. J. Integr. Med.* 15 (2), 93–94.
- Tap, J., Derrien, M., Törnblom, H., Brazeilles, R., Cools-Portier, S., Doré, J., Störsrud, S., Nevé, B.L., Öhman, L., Simrén, M., 2017. Identification of an intestinal microbiota signature associated with severity of irritable bowel syndrome. *Gastroenterology* 152 (1), 111–123.
- Toyokawa, Y., Takagi, T., Uchiyama, K., Mizushima, K., Inoue, K., Ushiroda, C., Kashiwagi, S., Nakano, T., Hotta, Y., Tanaka, M., Dohi, O., Okayama, T., Yoshida, N., Katada, K., Kamada, K., Ishikawa, T., Handa, O., Konishi, H., Naito, Y., Itoh, Y., 2019. Ginsenoside Rb1 promotes intestinal epithelial wound healing through extracellular signal-regulated kinase and Rho signaling. *J. Gastroenterol. Hepatol.* 34 (7), 1193–1200.
- Ueda, T., Shikano, M., Kamiya, T., Joh, T., Ugawa, S., 2011. The TRPV4 channel is a novel regulator of intracellular Ca²⁺ in human esophageal epithelial cells. *Am. J. Physiol. Gastrointest. Liver Physiol.* 301 (1), G138–G147.
- Verne, G.N., Sen, A., Price, D.D., 2005. Intrarectal lidocaine is an effective treatment for abdominal pain associated with diarrhea-predominant irritable bowel syndrome. *J. Pain* 6 (8), 493–496.
- Vriens, J., Owsianik, G., Voets, T., Droogmans, G., Nilius, B., 2004. Invertebrate TRP proteins as functional models for mammalian channels. *Pflug. Arch. Eur. J. Phy.* 449 (3), 213–226.
- Wang, J., Guo, T., Peng, Q.S., Yue, S.W., Wang, S.X., 2015. Berberine via suppression of transient receptor potential vanilloid 4 channel improves vascular stiffness in mice. *J. Cell Mol. Med.* 19 (11), 2607–2616.
- Wang, J., Ou, S.W., Bai, Y.F., Wang, Y.J., Xu, Z.Q.D., Luan, G.M., 2018. Downregulation of adult and neonatal Nav1.5 in the dorsal root ganglia and axon of peripheral sensory neurons of rats with spared nerve injury. *Int. J. Mol. Med.* 41 (4), 2225–2232.
- Wang, X., Xie, L., Long, J., Liu, K., Lu, J., Liang, Y., Cao, Y., Dai, X., Li, X., 2022. Therapeutic effect of baicalin on inflammatory bowel disease: a review. *J. Ethnopharmacol.* 283, 114749.
- Wang, Z., Ling, D., Wu, C., Han, J., Zhao, Y., 2020. Baicalin prevents the up-regulation of TRPV1 in dorsal root ganglion and attenuates chronic neuropathic pain. *Vet. Med. Sci.* 6 (4), 1034–1040.
- Wu, C.Y., Lin, Y.S., Yang, Y.H., Shu, L.H., Cheng, Y.C., Te Liu, H., 2020. Potential simultaneous inhibitors of angiotensin-converting enzyme 2 and transmembrane protease, serine 2. *Front. Pharmacol.* 11, 584158.
- Wu, D., Jiang, L., Wu, H., Wang, S., Zheng, S., Yang, J., Liu, Y., Ren, J., Chen, X., 2013. Inhibitory effects of glycyrrhetic acid on the delayed rectifier potassium current in Guinea pig ventricular myocytes and HERG channel. *Evid. Based Compl. Alternat. Med.* 2013, 481830.
- Wu, J., Nakashima, S., Shigyo, M., Yamasaki, M., Ikuno, S., Morikawa, A., 2020. Antihypertensive constituents in *Sanohashinto*. *J. Nat. Med.* 74, 421–433.
- Xu, G., 2006. Treatment of reflux laryngopharyngitis with modified *Banxia Xiexin tang* (*Pinellia* decoction for draining the heart)—a report of 40 cases. *J. Tradit. Chin. Med.* 26 (2), 127–131.
- Yennurajalingam, S., Tannir, N.M., Williams, J.L., Lu, Z., Hess, K.R., Frisbee-Hume, S., House, H.L., Lim, Z.D., Lim, K.H., Lopez, G., Reddy, A., Azhar, A., Wong, A., Patel, S. M., Kuban, D.A., Kaseb, A.O., Cohen, L., Bruera, E., 2017. A double-blind, randomized, placebo-controlled trial of *Panax ginseng* for cancer-related fatigue in patients with advanced cancer. *J. Natl. Compr. Cancer Netw.* 15, 1111–1120.
- Yu, S., Chun, E., Ji, Y., Lee, Y.J., Jin, M., 2021. Effects of red ginseng on gut microbiota and brain in a mouse model of post-infectious irritable bowel syndrome. *J. Ginseng Res.* 45 (6), 706–716.
- Yu, Y., Daly, D.M., Adam, I.J., Kitsanta, P., Hill, C.J., Wild, J., Shorthouse, A., Grundy, D., Jiang, W., 2016. Interplay between mast cells, enterochromaffin cells, and sensory signaling in the aging human bowel. *Neuro Gastroenterol. Motil.* 28 (10), 1465–1479.
- Yu, Z.C., Cen, Y.X., Wu, B.H., Wei, C., Xiong, F., Li, D.F., Liu, T.T., Luo, M.H., Guo, L.L., Li, Y.X., Wang, L.S., Wang, J.Y., Yao, J., 2019. Berberine prevents stress-induced gut inflammation and visceral hypersensitivity and reduces intestinal motility in rats. *World J. Gastroenterol.* 25 (29), 3956–3971.
- Yue, S.J., Qin, Y.F., Kang, A., Tao, H.J., Zhou, G.S., Chen, Y.Y., Jiang, J.Q., Tang, Y.P., Duan, J.A., 2021. Total flavonoids of *Glycyrrhiza uralensis* alleviates irinotecan-induced colitis via modification of gut microbiota and fecal metabolism. *Front. Immunol.* 12, 628358.
- Yue, Y., Wu, S., Li, Z., Li, X., Xiang, J., Ding, H., 2015. Wild jujube polysaccharides protect against experimental inflammatory bowel disease by enabling enhanced intestinal barrier function. *Food Funct.* 6 (8), 2568–2577.
- Zhang, C., Huang, Y., Li, P., Chen, X., Liu, F., Hou, Q., 2020. Ginger relieves intestinal hypersensitivity of diarrhea predominant irritable bowel syndrome by inhibiting proinflammatory reaction. *BMC Compl. Med. Ther.* 20, 279.

- Zhang, M., Ruwe, D., Saffari, R., Kravchenko, M., Zhang, W., 2020. Effects of TRPV1 activation by capsaicin and endogenous N-Arachidonoyl Taurine on synaptic transmission in the prefrontal cortex. *Front. Neurosci.* 14, 91.
- Zhang, W., Jiang, S., Qian, D., Shang, E., Duan, J., 2014. Effect of liquiritin on human intestinal bacteria growth: metabolism and modulation. *Biomed. Chromatogr.* 28 (9), 1271–1277.
- Zhao, T., Tang, H., Xie, L., Zheng, Y., Ma, Z., Sun, Q., Li, X., 2019. *Scutellaria baicalensis* Georgi. (Lamiaceae): a review of its traditional uses, botany, phytochemistry, pharmacology and toxicology. *J. Pharm. Pharmacol.* 71 (9), 1353–1369.
- Zheng, S., Xue, T., Wang, B., Guo, H., Liu, Q., 2022. Application of network pharmacology in the study of the mechanism of action of traditional Chinese medicine in the treatment of COVID-19. *Front. Pharmacol.* 13, 926901.
- Zhu, G., Ma, S., Li, X., Zhang, P., Tang, L., Cao, L., Liu, A., Sugita, T., Tomoda, T., 2018. The effect of ethanol extract of *Glycyrrhiza uralensis* on the voltage-gated sodium channel subtype 1.4. *J. Pharmacol. Sci.* 136 (2), 57–65.
- Zhu, Y., Huang, J., Shen, T., Yue, R., 2022. Mechanism of jujube (*Ziziphus jujuba* Mill.) fruit in the appetite regulation based on network pharmacology and molecular docking method. *Contrast Media Mol. Imaging* 2022, 5070086.
- Zielińska, M., Jarmuż, A., Wasilewski, A., Sałaga, M., Fichna, J., 2015. Role of transient receptor potential channels in intestinal inflammation and visceral pain: novel targets in inflammatory bowel diseases. *Inflamm. Bowel Dis.* 21 (2), 419–427.
- Zou, B., Cao, C., Fu, Y., Pan, D., Wang, W., Kong, L., 2022. Berberine alleviates gastroesophageal reflux-induced airway hyperresponsiveness in a transient receptor potential A1-dependent manner. *Evid. Based Compl. Alternat. Med.* 2022, 7464147.